MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING
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Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE, New Delhi
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Machine Learning Operations (23AIPE312)

Question Bank – Modules 1 to 5

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About this Document

This question bank has been prepared to help students prepare for the Continuous Internal Evaluation and the Semester End Examination in Machine Learning Operations. The questions and answers are framed by the course faculty, strictly in line with the prescribed syllabus and the textbook *Introducing MLOps* (Treveil et al., O'Reilly, 2020). AI assistance was used for the formatting and layout of the document.

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Syllabus

Course Title	Machine Learning Operations	**Course Code**	23AIPE312
Semester	V	**CIE Marks**	50
Teaching Hrs/Week (L:T:P)	2:0:2	**SEE Marks**	50
Total Hrs	26 + 24 = 50	**Exam Hrs**	03
Credits	03		

Course Learning Objectives: This course is designed to

1. Impart the fundamental concepts of Machine Learning Operations and its role in model selection.
2. Process in different machine learning model development, versioning, and reproducibility principle.
3. Familiarize best practices for deploying machine learning models using modern DevOps tools.

Module	Topics	Hrs	Textbook 1
Module 1: Introduction to MLOps	Defining MLOps and Its Challenges, MLOps to Mitigate Risk, MLOps for Scale, People of MLOps, Subject Matter Experts, Key MLOps Features, A Primer on Machine Learning, Model Development, Productionalization and Deployment, Monitoring, Iteration and Life Cycle, Governance.	5+2	Part I (1, 2, 3)
Module 2: Process in Model Development and Versioning	Machine Learning Model, ML Required Components, Different ML Algorithms and Different MLOps Challenges, Data Exploration, Feature Engineering and Selection, Experimentation, Evaluating and Comparing Models, Version Management and Reproducibility.	5+4	Part II (4)
Module 3: Preparing for Production & CI/CD Pipelining	Runtime environments, Model risk evaluation, Quality assurance for ML, Key testing consideration, Reproducibility & Auditability, ML security, Model risk mitigation, Deploying to production: CI/CD Pipelines for ML, Building ML Artifacts, Deployment strategies, Containerization, Scaling Deployments, Requirements & Challenges.	5+4	5, 6
Module 4: Monitoring, Feedback Loop & Governance	Model retrainment, Understanding Model Degradation, Drift detection in practice, Feedback loop, Model Governance, Matching governance with risk level, Current Regulations Driving MLOps Governance, The New Wave of AI-Specific Regulations and Emergence of Responsible AI.	6+4	6, 7, 8

Module	Topics	Hrs	Textbook 1
Module 5: MLOps in Practice	Consumer credit risk management: Model development, Model bias consideration, prepare & deploy for production. Marketing Recommendation Engines: Data preparation, Design & manage experiment, Model training & Deployment. Consumption forecast: Power Systems, Data Collections.	5+4	9, 10, 11

Laboratory Experiments:

1. **Experiment 1:** Set up GitHub/GitLab for ML project versioning. Familiar with Git commands. Use git for tracking model and code changes.
2. **Experiment 2: Experiment Tracking with MLflow.** Train a simple ML model (e.g., logistic regression). Log parameters, metrics, and artifacts using MLflow.
3. **Experiment 3: Implementing CI/CD for ML Models.** Set up a CI/CD pipeline using GitHub Actions. Automate model training and testing on code commits.
4. **Experiment 4: Containerizing the Model.** Train a simple ML model (e.g., a classification model using scikit-learn). Save the model using joblib or pickle. Write a Flask or FastAPI app to serve the model as an API. Create a Dockerfile to containerize the model. Build and run the Docker container.
5. **Experiment 5:** (a) **Developing a Credit Risk Model:** use a dataset like German Credit Risk Dataset or Home Credit Default Risk; train a classification model (Logistic Regression, Random Forest, or XGBoost) to predict loan default probability; track the model lifecycle using MLflow (versioning, experiment tracking). (b) **Creating a Continual Learning Model:** train a recommendation engine using the MovieLens dataset; set up an online learning pipeline that updates the model with new user interactions.
6. **Mini Project** (6 Hrs).

Course Outcomes: At the end of the course, the student will be able to

1. **CO1:** Understand the fundamentals of MLOps, ML model deployment and its significance.
2. **CO2:** Compare various model versioning techniques and experiment tracking methods.
3. **CO3:** Implement model deployment strategies using Docker, Kubernetes, and cloud-based solutions.
4. **CO4:** Apply ML models in production, detecting performance degradation and drift.
5. **CO5:** Develop an end-to-end MLOps workflow by integrating best practices in automation and testing.

Textbooks:

1. Mark Treveil, Nicolas Omont, Clément Stenac, Kenji Lefevre, Du Phan, Joachim Zentici, Adrien Lavoillotte, Makoto Miyazaki, Lynn Heidmann, *Introducing MLOps:*

How to Scale Machine Learning in the Enterprise, Sebastopol, CA, USA: O'Reilly Media, 2020.

2. Emmanuel Raj, *Engineering MLOps: Rapidly build, test, and manage production-ready machine learning life cycles at scale*, Packt Publishing, 2021.

Reference Books:

1. Saurabh D. Dorle, *Ultimate MLOps for Machine Learning Models*, AVA Publishing, 2024.
2. Mason Leblanc, *MLOps Mastery: Building and Scaling Trustworthy AI Systems and Robust ML Pipelines with Machine Learning Operations*, 2024.

Web Links:

1. freeCodeCamp course — MLOps Course: Build Machine Learning Production Grade Projects: https://www.youtube.com/watch?v=-dJPoLm_gtE
2. GitHub MLOps-course: https://github.com/GokuMohandas/mlops-course
3. GitHub Awesome-MLOps-Course: https://github.com/ashishpatel26/Awesome-MLOps-Course

Scope of the Question Bank

84 questions, each mapped to a syllabus topic and a page of Textbook 1: 16 per module (8 of 8 marks and 8 of 6 marks), plus 2 application questions (6 marks, RBT level L3) each in Modules 4 and 5. The application questions apply the textbook content to a scenario.

Module	Textbook 1 chapters used
Module 1	Chapter 1 (pp. 3–11), Chapter 3 (pp. 23–38)
Module 2	Chapter 4 (pp. 41–58)
Module 3	Chapter 5 (pp. 59–72), Chapter 6 (pp. 73–84)
Module 4	Chapter 7 (pp. 85–103), Chapter 8 (pp. 105–113)
Module 5	Chapter 9 (pp. 129–133), Chapter 10 (pp. 135–141), Chapter 11 (pp. 147–151)

- **Chapter 6:** the syllabus lists Textbook 1 Chapter 6 under Module 4, but its topics (CI/CD pipelines, ML artifacts, deployment strategies, containerization, scaling deployments) are named in Module 3, so it is treated as Module 3.
- **Sections outside the syllabus** are not used: Chapter 8 from "Key Elements of Responsible AI" onwards, Chapter 10 from "Pipeline Structure and Deployment Strategy" onwards, and Chapter 11 from "Problem Definition" onwards.
- **Not covered:** Chapter 2, *People of MLOps* (pp. 13–22: Subject Matter Experts, Data Scientists, Data Engineers, Software Engineers, DevOps, Model Risk Manager/Auditor, ML Architect).
- **Laboratory experiments** are not included; this bank covers theory questions only.
- **Answer length guide:** an 8-mark answer expects about 8 distinct points or a point plus explanation per mark band; a 6-mark answer expects about 6.

Module 1

Introduction to MLOps

Module 1 – 8 Mark Questions

Q1. Define MLOps. Explain the three key reasons why managing machine learning life cycles at scale is challenging. (8 marks)

Topic: Defining MLOps and Its Challenges · Ch. 1, pp. 4–5

Definition (2 marks): MLOps is the standardization and streamlining of machine learning life cycle management. It helps organizations generate long-term value and reduce the risk of data science, ML and AI initiatives. With decision automation (decisions made without human intervention), models become more critical and managing their risk becomes a top-level concern.

Three reasons (6 marks):

1. **Many dependencies.** Data changes constantly and business needs shift too. Results must be relayed back to the business continually, so that the model's behaviour on production data still matches expectations and still addresses the original goal.
2. **Not everyone speaks the same language.** The life cycle involves business, data science and IT teams. These groups use different tools and often lack shared fundamental skills, so there is no common baseline for communication.
3. **Data scientists are not software engineers.** Most specialize in model building and assessment, not in writing applications. Many juggle several roles and are stretched thin, which becomes serious at scale with more models to manage. Staff turnover makes it worse: data scientists end up managing models they did not create.

The simple life cycle (Fig. 1-2) understates this; the realistic picture (Fig. 1-3) involves many people with different skills and tools.

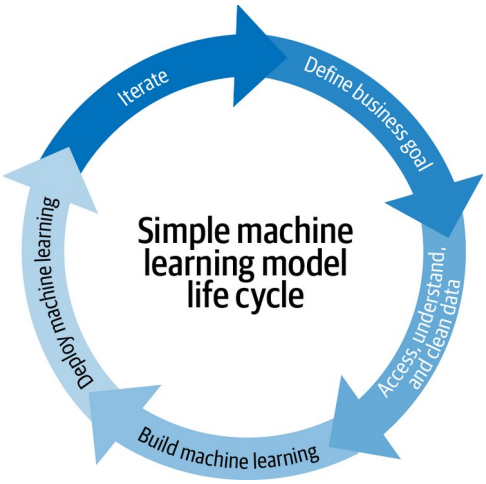


Figure 1-2. A simple representation of the machine learning model life cycle (Source: Treveil et al., Introducing MLOps, 2020)

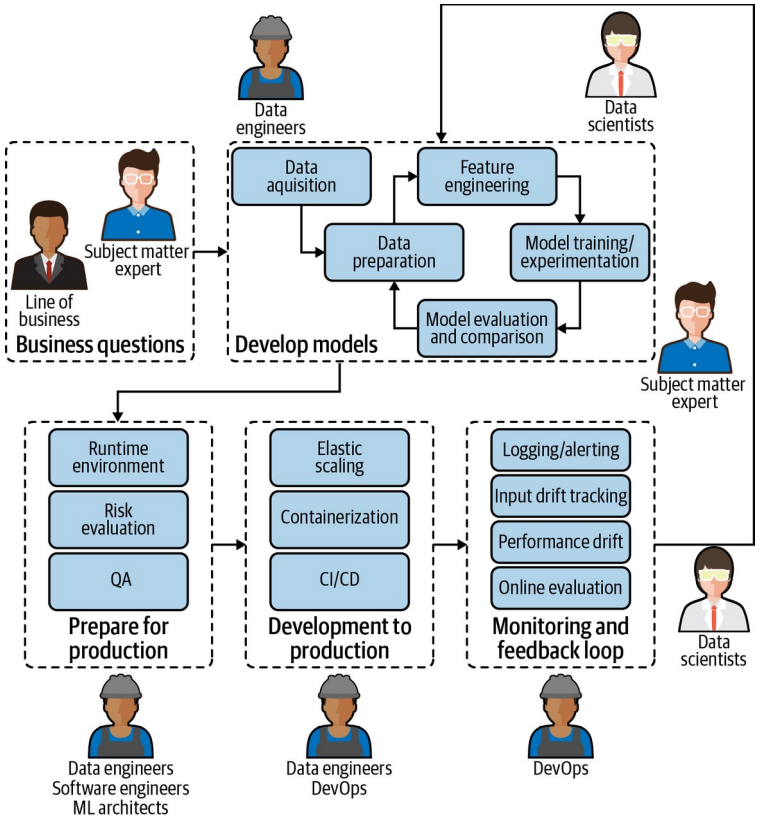


Figure 1-3. The realistic picture of a machine learning model life cycle inside an average organization (Source: Treveil et al., Introducing MLOps, 2020)

Q2. Compare MLOps with DevOps. How do DataOps, ModelOps and AIOps relate to MLOps? (8 marks)

Topic: Defining MLOps and Its Challenges · Ch. 1, pp. 3, 6–7

What MLOps and DevOps share (2 marks):

- Robust automation and trust between teams
- Collaboration and increased communication between teams
- The end-to-end service life cycle (build, test, release)
- Priority on continuous delivery and high quality

The critical difference (2 marks): Deploying software code is fundamentally different from deploying ML models. Software code is relatively static, but data is always changing, so models must keep learning and adapting to new inputs. Because an ML model is made of both code and data, MLOps is a new and unique discipline.

Term	Meaning	Relation to MLOps
ModelOps	Emerged with MLOps in 2018–2019; argued to be more general, covering any model (e.g., rule-based)	Largely used interchangeably; the textbook uses “MLOps” for the ML life cycle

Term	Meaning	Relation to MLOps
AIOps	Using AI to solve operational challenges (AI for DevOps), e.g., predictive maintenance for network failures	A different topic entirely, outside MLOps
DataOps	Introduced by IBM in 2014; delivers business-ready data quickly, focusing on data quality and metadata management	Intersects with MLOps, but MLOps adds more robustness through its key features

DataOps example (2 marks): If data a model relies on changes suddenly, a DataOps system alerts the business team to treat recent insights carefully and notifies the data team to investigate or roll back a library upgrade and rebuild the affected partition.

Why now (2 marks): Teams managed without centralized processes because they were not deploying models at large enough scale. They now need a multi-stage, multi-discipline, multi-phase process in a heterogeneous environment.

Q3. Explain how MLOps helps to mitigate risk, covering risk assessment and risk mitigation. (8 marks)

Topic: MLOps to Mitigate Risk · Ch. 1, pp. 7–9

MLOps matters to any team with even one model in production, because continuous performance monitoring and adjustment may be essential. MLOps has a cost, so a cost-benefit evaluation is done per use case.

Risk assessment (4 marks): Risks vary widely; a monthly marketing-offer recommender carries far less risk than a travel site whose pricing depends on a model. The analysis should cover the risk that:

1. The model is unavailable for a period of time
2. The model returns a bad prediction for a sample
3. Model accuracy or fairness decreases over time
4. The skills needed to maintain the model (data science talent) are lost

Risk is larger for models deployed widely and used outside the organization. Assessment uses two metrics, the **probability** and the **impact** of the adverse event; their combination gives the model's **severity** (Fig. 1-4). It is done at project start and reassessed periodically.

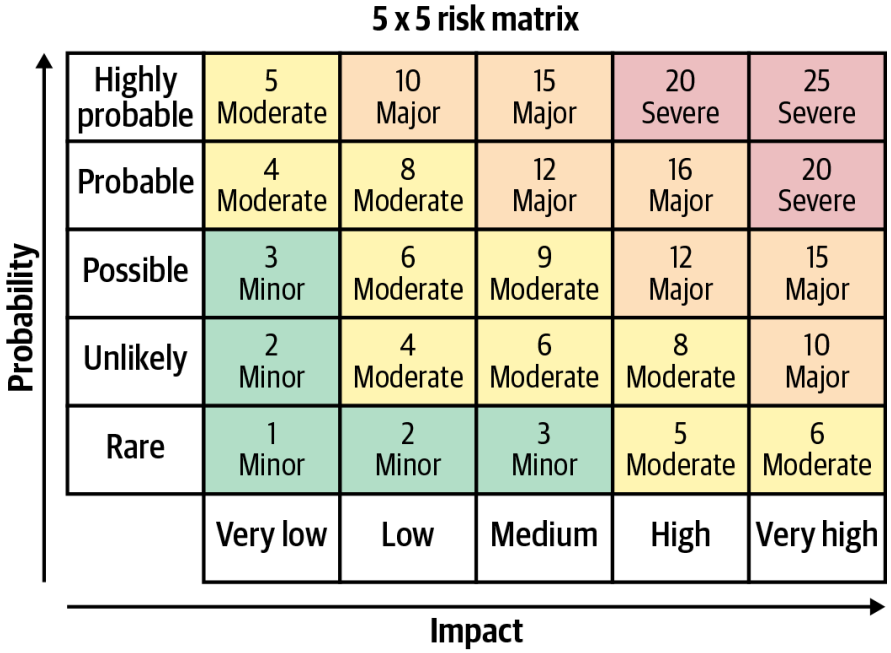


Figure 1-4. A table that helps decision makers with quantitative risk analysis (Source: Treveil et al., Introducing MLOps, 2020)

Risk mitigation (4 marks):

- MLOps becomes critical once a centralized team runs more than a handful of models; without standardization there is no global view of model states.
- A model's performance can often only be fully assessed in production, because training data must reflect production data. If production changes, performance can drop rapidly.
- Models are sensitive to their environment. They are built on open-source software (scikit-learn, Python, Linux), so production versions must match the versions the model was verified on.
- Deployment is the beginning, not the end: monitoring ensures expected behaviour and tells the organization how broadly each model is used.

Q4. Explain the stages of model development as a key feature of MLOps. (8 marks)

Topic: Model Development · Ch. 3, pp. 24–27

1. **Establishing business objectives.** Development starts with an objective, e.g., reducing fraudulent transactions below 0.1%. Objectives bring performance targets, infrastructure needs and cost constraints, captured as KPIs that later let production models be monitored. Objectives also include change management; e.g., the required transparency influences algorithm choice.
2. **Data sources and exploratory data analysis (EDA).** Subject matter experts and data scientists search for suitable input data, often the hardest part. EDA builds hypotheses, identifies cleaning needs and informs feature selection, visually or statistically.
3. **Feature engineering and selection.** Raw data is transformed into features (fixed-size arrays of numbers, the only object ML algorithms understand). It includes data cleansing, which can take the most project time.

4. **Training and evaluation.** Training is iterative: algorithms tested, features generated, selections adapted, hyperparameters tuned. It is the most compute-intensive step. An experiment tracking tool records data, features, parameters and metrics for side-by-side comparison. The best solution needs quantitative criteria (accuracy, error) and qualitative ones (explainability, ease of deployment).
5. **Reproducibility.** Significant model versions are saved with enough information about the environment to reproduce the same results from scratch. This needs version control of all assets, including data, and a record of the software environment; otherwise the model cannot be handed to DevOps reliably.
6. **Responsible AI.** DevOps must know how to verify the model; regulated industries document how it was built and tuned. Documentation (partly automated) and explainability techniques (partial dependence plots, subpopulation analyses, Shapley values, what-if analysis) address this.

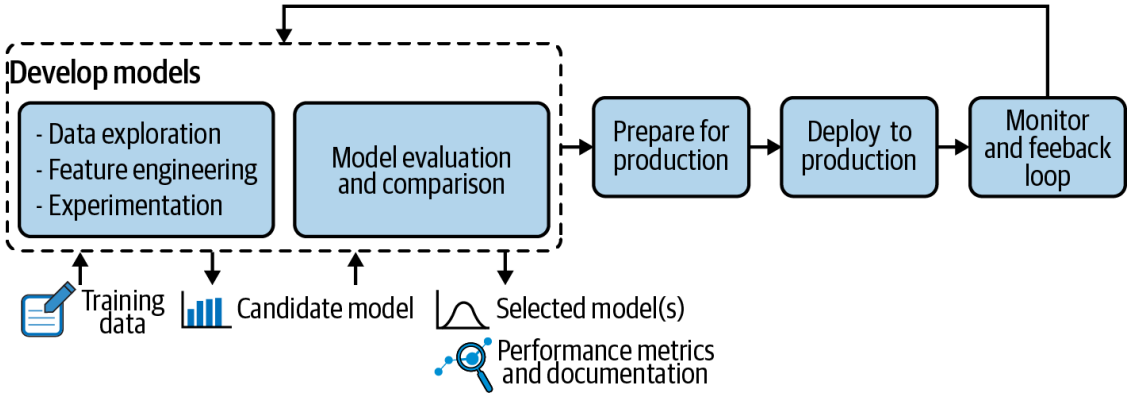


Figure 4-1. Model development highlighted in the larger context of the ML project life cycle (Source: Treveil et al., Introducing MLOps, 2020)

Q5. Explain productionalization and deployment of ML models: deployment types, model contents, containerization and deployment requirements. (8 marks)

Topic: Productionalization and Deployment · Ch. 3, pp. 27–29

Productionalization is the domain of software engineers and DevOps; poor collaboration with data scientists causes delays or failed deployments.

Deployment types (2 marks):

- **Model-as-a-service (live scoring):** the model sits in a simple framework exposing a REST API endpoint that answers requests in real time.
- **Embedded model:** the model is packaged into an application that is published, e.g., an application that batch-scores requests.

Model contents (1 mark): Code (Python, R or Java) plus data artifacts. Runtime and package versions must match production, or predictions may differ.

Portable formats (1 mark): PMML, PFA, ONNX or POJO reduce dependency on the production environment, but each supports limited algorithms and may behave subtly differently from the original.

Containerization (1 mark): Docker containers are lightweight alternatives to virtual machines, giving each model a self-contained environment. They enable blue-green deployment and elastic scaling; Kubernetes orchestrates many containers, in the cloud or on-premise.

Deployment requirements (3 marks): Rapid automated deployment is preferred. Short-lived self-service apps may use a single-step push to production, with resources capped by Linux cgroups and simple UIs in Flask. Customer-facing, mission-critical cases need a robust CI/CD pipeline:

1. Meet coding, documentation and sign-off standards
2. Re-create the model in a production-like environment
3. Revalidate model accuracy
4. Perform explainability checks
5. Meet all governance requirements
6. Check the quality of data artifacts
7. Test resource usage under load
8. Embed into the larger application with integration tests

Regulated industries (finance, pharmaceuticals) add manual checks; the aim is still to automate as far as possible for faster deployment, more regression testing and fewer errors.

Q6. Explain model monitoring from the perspectives of the DevOps team, the data scientist and the business. (8 marks)

Topic: Monitoring · Ch. 3, pp. 29–32

Good performance means different things to different people, so monitoring has three viewpoints.

DevOps concerns (2 marks): Is the model fast enough? Is it using sensible memory and processing time? This is traditional IT performance monitoring that DevOps already does well. Compute scalability matters, e.g., when retraining in production; deep learning needs more resources than decision trees.

Data scientist concerns (4 marks): Models degrade over time because they are models of the data they were trained on, and the real world changes. A fraud model trained six months ago misses a new fraud type; an ad model degrades as a site's audience gets younger. Retraining timing depends on how fast the world changes, required accuracy and how easy it is to deploy a better model. Degradation is detected two ways:

- **Ground truth:** the correct answer to the model's question (was this transaction really fraud?). It can arrive in seconds (ad clicks) or a month later (cardholder reports), too slow for daily monitoring.
- **Input drift:** compare recent requests against training data; distinct differences signal likely compromised performance. No waiting for ground truth is needed.

Business concerns (2 marks): Is the model delivering value? Do benefits outweigh development and deployment costs? The original KPIs should be monitored, automatically where possible. With no "dollar-o-meter", monitoring business KPIs against a sensible baseline (e.g., a rule-based model built on subject matter expertise) isolates the ML contribution.

Q7. Explain iteration and the life cycle of ML models, including retraining pitfalls and the feedback loop. (8 marks)

Topic: Iteration and Life Cycle · Ch. 3, pp. 32–34

Reasons for a new version (1 mark): performance degradation from drift, refined business objectives or KPIs, or a better model design.

Iteration (3 marks): In fast-moving settings new data arrives daily, and retraining and redeployment are automated. Even simple retraining has pitfalls:

- Does the new training data look as expected? Automated validation with predefined metrics is essential.
- Is the data complete and consistent?
- Are feature distributions similar to the previous training set? The goal is to refine, not radically change.

The new version is compared with the live model on the same development dataset. Wide variation means no automatic redeployment; manual intervention is needed. If drift is detected but little labelled data exists, data scientists must find the cause and adjust training data, e.g., remove a feature or resample rows.

Feedback loop (4 marks): Live scoring and retraining environments are kept separate, so evaluation in the retraining environment is uncertain.

- **Shadow testing:** the new model runs alongside the incumbent; the incumbent serves all requests, the new model scores them too and logs results without returning them. Results are compared statistically; it also gives SMEs visibility of future versions.
- **A/B testing:** requests are split; each is handled by one model only. Used when outcomes need user reaction (e.g., ad clicks). The fixed-horizon test waits for a predetermined sample size; peeking early is unreliable and waiting can be costly.
- **Multi-armed bandit (Bayesian):** adaptively shifts traffic away from underperforming models, reaching conclusions faster and lowering business cost.

Q8. Explain governance in MLOps: its definition, legal obligations, GDPR principles and the two categories of governance initiatives. (8 marks)

Topic: Governance · Ch. 3, pp. 34–38

Definition (1 mark): Governance is the set of controls that ensure a business delivers on its responsibilities to all stakeholders (shareholders, employees, the public, governments). These are financial, legal and ethical obligations, all underpinned by fairness.

Legal obligations (1 mark): Industry regulations long predate ML, e.g., finance rules against mismanagement and fraud, pharmaceutical rules for public health, and broader laws ensuring a level playing field on sex, race, age or religion.

Data privacy regulation (3 marks): The 2016 EU GDPR and the 2018 California CCPA protect the public from misuse of personal data, with large impact on data-dependent ML. GDPR principles for processing personal data:

1. Lawfulness, fairness and transparency
2. Purpose limitation
3. Data minimization
4. Accuracy
5. Storage limitation
6. Integrity and confidentiality (security)
7. Accountability

The EU is also planning AI-specific legislation. Beyond law, ESG indicators and public activism push businesses toward Responsible AI: ethical, transparent and accountable use of AI.

Two categories (3 marks):

- **Data governance:** ensures appropriate use and management of data — provenance, terms of collection, accuracy, and whether PII or sensitive data is present. Data lineage is hard but essential for GDPR compliance; anonymization alone may not prevent re-identification. Accidental bias is a risk, e.g., Amazon's recruiting tool that discriminated against women (scrapped in 2018).
- **Process governance:** well-defined process steps with reviews, sign-offs and documentation, so every consideration is made at the right time (no deployment before validation) and outsiders (auditors, risk managers, compliance officers) can review decisions later.

Module 1 – 6 Mark Questions

Q9. Explain the two dimensions of Responsible AI and the interplay between MLOps and Responsible AI. (6 marks)

Topic: MLOps to Mitigate Risk (MLOps for Responsible AI) · Ch. 1, pp. 9–10

Intentionality (2 marks): Models are designed and behave in line with their purpose. Data must come from compliant, unbiased sources, with collaborative checks and balances on model bias. It also includes explainability: results should be explainable by humans, not only the system's creators.

Accountability (2 marks): Central control, management and auditing of enterprise AI — no shadow IT. It means knowing which teams use what data, how, and in which models; trusting

that data is reliable and collected per regulations; and traceability, i.e., finding easily where in the pipeline something went wrong.

Interplay (2 marks): ML models lack the transparency of imperative code, so proving compliance is harder. Automation moves accountability from individual contributors (e.g., pricing or loan decisions) up to data team managers or executives. Good MLOps is needed to practise Responsible AI, and Responsible AI requires MLOps strategies.

Q10. Explain why MLOps is essential for scale. (6 marks)

Topic: MLOps for Scale · Ch. 1, pp. 10–11

- **Economies of scale (1 mark):** Moving from one or a few models to tens, hundreds or thousands with positive business impact requires MLOps discipline.
- **Minimum benefits of good practice (3 marks):**
  1. Track versioning, especially of experiments in the design phase
  2. Know whether retrained models beat previous versions and promote the better ones to production
  3. Ensure at defined periods (daily, monthly) that production performance is not degrading
- **Not optional (1 mark):** Without MLOps, teams face problems of model quality and continuity, or deploy models with real negative impact, such as biased predictions that hurt the company's reputation.
- **Transparency (1 mark):** Upper management and the C-suite should see which models are in production, their business effect, and the full data pipeline from raw data to output. MLOps provides this transparency and accountability.

Q11. Differentiate between ground truth and input drift as approaches to detect model performance degradation. (6 marks)

Topic: Monitoring · Ch. 3, pp. 30–31

Models degrade because they represent the data they were trained on, and the real world changes.

Aspect	Ground truth	Input drift
Meaning	The correct answer to the question the model was asked (e.g., was the transaction actually fraudulent?)	A difference between recent requests to the model and the training data
Principle	Knowing the true answer for each prediction shows how well the model performs	A model predicts accurately only if training data reflects the real world
Speed	Fast for ad clicks (seconds); slow for fraud (phone call in minutes, cardholder report up to a month, or never)	Immediate; all needed data already exists

Aspect	Ground truth	Input drift
Waiting needed	Yes, for labels to arrive	No
Best when	Feedback comes quickly	Rapid feedback is required

Identifying drift is one of the most important parts of an adaptable MLOps strategy and brings agility to enterprise AI.

Q12. Explain shadow testing and A/B testing in the model feedback loop. (6 marks)

Topic: Iteration and Life Cycle (The Feedback Loop) · Ch. 3, pp. 33–34

Need (1 mark): Live scoring and retraining environments are separate in large enterprises, so evaluating a new version in the retraining environment is uncertain.

Shadow testing (2 marks): The new version is deployed in the live environment beside the existing model. The incumbent serves every request; the new model scores the same request, and its result is logged but not returned. After enough requests, results are compared statistically. It also gives SMEs visibility of future versions for a smoother transition.

A/B testing (2 marks): Both models are live, but requests are split; each request goes to one model only. It suits cases like ad selection, where a prediction's quality is known only after the user reacts. Its simplest form, the fixed-horizon test, must run to a predetermined sample size; peeking early is unreliable, and not stopping early can be costly.

Multi-armed bandit (1 mark): A Bayesian, adaptive alternative that shifts traffic away from underperforming models, reaching conclusions faster and reducing business cost.

Q13. Explain the challenges of iterating ML models on the edge with suitable examples. (6 marks)

Topic: Iteration and Life Cycle (Iterating on the Edge) · Ch. 3, p. 34

Challenge (1 mark): Models deployed on millions of devices (smartphones, sensors, cars) face different iteration challenges from corporate IT.

Centralized retraining – Tesla Autopilot (2 marks): Feedback from all model instances is relayed to a central point and training happens centrally. Autopilot runs in over 500,000 cars; fully retraining its roughly 50 neural networks takes 70,000 GPU hours.

Federated learning – Google GBoard (2 marks): Each smartphone retrains the keyboard model locally and sends only a summary of improvements to Google. Improvements from all devices are averaged to update the shared model.

Benefits of federated learning (1 mark): Personal data need not be collected centrally, each phone can use its improved model immediately, and overall power consumption falls.

Q14. Differentiate between data governance and process governance in MLOps. (6 marks)

Topic: Governance · Ch. 3, pp. 36–38

Aspect	Data governance	Process governance
Definition	A framework ensuring appropriate use and management of data	Well-defined processes ensuring governance is addressed at the right point in the model life cycle, with a full record kept
Focus	Data used, especially for training	Formal steps of the MLOps process and actions tied to them
Key questions / actions	Data provenance; how and under what terms it was collected; accuracy and currency; presence of PII or sensitive data	Reviews, sign-offs, capture of documentation
Aim	Lineage: where data came from and the downstream effect of changing it; GDPR compliance; avoiding accidental bias	Right decision at the right time (no deployment before validation); oversight by auditors, risk managers, compliance officers
Difficulties	Lineage is hard at feature level; anonymization may still allow re-identification; tools are in their infancy	Process knowledge is spread across teams; every team must adopt it; a heavy process gets subverted
Where common	Needed wherever personal data is used	Mostly in heavily regulated sectors such as finance

Example of data governance failure: an ML recruitment model learned bias against women because all-female schools were under-represented in upper management (Amazon, 2018).

Q15. What key questions must be answered when finding data to build ML models, including data governance constraints? (6 marks)

Topic: Model Development (Data Sources and EDA) · Ch. 3, pp. 24–25

Subject matter experts and data scientists search for input data together; this is often the most arduous part.

Questions for finding data (4 marks):

- What relevant datasets are available, and are they accurate and reliable?
- How can stakeholders access the data?
- What features can be created by combining sources?
- Will the data be available in real time?
- Must data be labelled with ground truth, or is unsupervised learning suitable, and at what cost?
- Which platform should be used?
- How will data be updated after deployment, and will using the model reduce the data’s representativeness?
- How will the business KPIs be measured?

Data governance questions (2 marks):

- Can the datasets be used for this purpose, and under what terms of use?
- Is there PII that must be redacted or anonymized?
- Are there features, such as gender, that legally cannot be used here?
- Are minority populations represented well enough for equivalent performance on each group?

EDA then builds hypotheses, finds cleaning needs and guides feature selection.

Q16. Explain the role of documentation and explainability in Responsible AI during model development. (6 marks)

Topic: Model Development (Responsible AI) · Ch. 3, pp. 26–27

Verification and documentation (2 marks): Reproducing a model is not enough; DevOps must know what the model does, how to test it and the expected results. Regulated industries document how it was built and tuned, and critical models may be independently recoded. Automated model documentation helps, but some choices must be explained by hand.

Why explainability (1 mark): Performance measures do not explain how predictions are made. Knowing the "how" helps sanity-check the model, engineer better features and confirm fairness on features like sex, age or race.

Techniques (3 marks):

1. **Partial dependence plots:** the marginal impact of features on the predicted outcome
2. **Subpopulation analyses:** how the model treats specific subpopulations; the basis of many fairness analyses
3. **Individual model predictions (e.g., Shapley values):** how each feature value contributes to a specific prediction
4. **What-if analysis:** the sensitivity of a prediction to its inputs

Module 2

Process in Model Development and Versioning

Module 2 – 8 Mark Questions

Q17. What is a machine learning model? Explain it in theory and in practice with examples. (8 marks)

Topic: Machine Learning Model · Ch. 4, pp. 42–44

In theory (4 marks):

- A model is a projection of reality: a partial, approximate representation of some aspects of a real thing or process, chosen by what is available and useful.
- Once trained, it reduces to a mathematical formula that returns a result for given inputs, such as the probability of an event or an estimated value.
- Models rest on statistical theory; ML algorithms are the tools that build models from training data by finding a synthetic representation of it. Since data reflects the world at collection time, models predict well when the future resembles the past.
- **Generalization capacity** is the ability to predict accurately for cases unlike the input data, e.g., CycleGAN producing zebra-striped horses never seen in training.
- Examples: house price prediction from surface area, bedrooms, location and market state; disease prediction, a classification model that outputs a probability, sometimes with a confidence interval.

In practice (4 marks):

- A model is the set of parameters needed to rebuild and apply the formula. It is usually stateless and deterministic: the same inputs give the same outputs.
- It includes every transformation from raw input to the final formula and derived outputs (a class or decision). Whether ML-based or not, it is a computable function applied one row at a time.
- House price example: zip codes may be replaced by derived inputs such as average income, population density or proximity to amenities. Users still enter the zip code, so that transformation is part of the model.
- Outputs can be richer than one number: a fraud system returns a probability and flags fraud only above a fine-tuned threshold; some models output recommendations or decisions.
- These parts need not be one monolithic artifact, since they have different refresh rates and external sources.

Q18. Explain the required components of a machine learning model with suitable examples. (8 marks)

Topic: ML Required Components · Ch. 4, p. 44 (Table 4-1)

1. **Training data (2 marks):** Usually labelled with examples of what is modelled (supervised learning). Quality is vital. Example: World War II data from damaged planes that returned suffered from survivor bias, making it poor training data.

2. **Performance metric (2 marks):** What the model optimizes. A careless choice causes unintended consequences, like the cobra effect (a bounty on dead cobras led people to breed them). If 95% of data is class A, optimizing raw accuracy can yield a model that always predicts A and is 95% accurate yet useless.
3. **ML algorithm (1 mark):** Many algorithms exist with different pros and cons; choice depends on the task and priorities such as performance, stability, interpretability and computation cost.
4. **Hyperparameters (2 marks):** Configurations of the algorithm. The algorithm holds the basic formula, the learned parameters are the operations and operands for this task, and hyperparameters govern how the algorithm finds those parameters. Example: tree depth (number of splits) in a decision tree.
5. **Evaluation dataset (1 mark):** With labelled data, a dataset separate from training data measures performance on unseen data, i.e., generalization.

The number and complexity of these components, plus algorithm choice, make good MLOps challenging.

Q19. "Different ML algorithms present different MLOps challenges." Justify this statement. Also explain the role of computing power. (8 marks)

Topic: Different ML Algorithms and Different MLOps Challenges · Ch. 4, pp. 45–46 (Table 4-2)

All ML algorithms model patterns in past data; the quality and relevance of that experience decide their effectiveness. Each algorithm type has its own MLOps considerations (5 marks):

Type	Algorithm	MLOps consideration
Linear	Linear regression	Tends to overfit
Linear	Logistic regression	Tends to overfit
Tree-based	Decision tree	Unstable: small data changes can greatly change the optimal tree
Tree-based	Random forest	Predictions hard to understand (a Responsible AI concern); relatively slow to predict
Tree-based	Gradient boosting	Hard to understand; small changes in features or training set can radically change the model
Deep learning	Neural networks	Almost impossible to interpret; very slow to train; need much power and data — is a simpler model enough?

Governance and algorithm choice (1 mark): Highly regulated settings that must explain decisions (e.g., financial services) cannot use opaque models like neural networks and favour simpler techniques such as decision trees. The trade-off is often cost, not performance: simpler techniques need costlier manual feature engineering for equal performance.

Computing power (2 marks): ML development is inversely proportional to the cost of computing power. New algorithms appeared as power became affordable, from hand-

computed linear regression to today's deep learning. Random forest and gradient boosting were expensive 20 years ago but lowered the cost of building models. Cheaper data helped less: most algorithms still keep all training data in memory rather than distributing it.

Q20. Explain feature engineering with its categories and techniques. Add a note on transfer learning. (8 marks)

Topic: Feature Engineering and Selection · Ch. 4, pp. 47–48

Definition (1 mark): Features are how data is presented to a model, telling it things it may not infer by itself.

Categories with examples (3 marks):

Category	Description	Example
Derivatives	Infer new information from existing information	Day of the week from a date
Enrichment	Add external information	Is this day a public holiday?
Encoding	Present the same information differently	Weekday versus weekend
Combination	Link features together	Backlog size weighted by item complexity

Business-process example: from a date, derive the weekday or distance to the next public holiday, and consider each location's calendar. House-price example: use average income and population density instead of segmenting by zip code.

Techniques (3 marks):

- **Data enrichment services:** a market for complementary data beyond public open data.
- **Impact coding:** replace a category by the average target value for that category, at some information loss.
- **One-hot encoding:** a feature with three values (Raspberry, Blueberry, Strawberry) becomes three yes/no features.
- **Embeddings:** deep learning turns text and images into tables of numbers usable by ML algorithms.

Transfer learning (1 mark): Using knowledge gained on one problem to solve another, accelerating later tasks; popular in deep learning. Example: a model never trained on forks can still provide useful embeddings for a fork detector, because it learned similar human-made objects.

Q21. Explain experimentation in model development, including its goals, bias and variance, and how it is managed. (8 marks)

Topic: Experimentation · Ch. 4, pp. 49–51

Nature (1 mark): Experimentation runs through all of model development; important decisions are justified by experiments, ranging from full predictive models to statistical tests or charts.

Goals (2 marks):

1. Assess how useful a model can be built from the required components (Table 4-1)
2. Find the best modelling parameters (algorithms, hyperparameters, feature preprocessing)
3. Tune the bias/variance trade-off for a given training cost
4. Balance model improvement against computation cost — how good is good enough?

Bias and variance (2 marks):

- **High bias (underfitting):** misses rules learnable from data, often due to reductive, oversimplified assumptions.
- **High variance (overfitting):** treats noise as patterns, predicting every variation; complex and poor at generalizing.

Managing experiments (3 marks):

- Tools run experiments semi-automatically: you define the space of possibilities from prior knowledge and constraints (computation, budget).
- **Stratified modelling:** e.g., one demand model per store instead of one for all stores, since behaviour varies by store.
- Trying every combination is intractable, so set a time and/or computation budget and an acceptability threshold.
- Experiments may need rerunning whenever data or constraints change significantly; platforms preserve processing steps for repeatability and support version control and experimental branches to merge, discard or keep.

Q22. Explain how ML models are evaluated and compared, including choice of metrics and cross-testing. (8 marks)

Topic: Evaluating and Comparing Models · Ch. 4, pp. 51–53

Philosophy (2 marks): George Box noted that all models are wrong but some are useful. A model must be good enough to be useful, while avoiding the uncanny valley: seemingly good but catastrophic for a subset, e.g., an under-represented population. Evaluate in context against what existed before (a previous model or rules). A modest model can still save a lot, e.g., slightly better demand forecasts.

Suspicious perfect scores (1 mark): Most data has noise, so a near-perfect score suggests data leakage (target in inputs, or a feature available only after the target is known) or overfitting.

Choosing metrics (2 marks): The metric shapes the model (cobra effect). Accuracy suits balanced classification but fails for imbalanced classes: if the positive class is 5%, always

predicting negative is 95% accurate and useless. No metric fits all; understand its maths trade-offs and business impact.

Cross-testing (2 marks): Evaluate on a holdout set not used in training. Data can be held out at several steps (metric evaluation, hyperparameter optimization). **k-fold cross-validation** rotates the held-out parts across several trainings, multiplying training time but showing metric stability. A holdout of the most recent records gives more realistic estimates and reveals drift between training and holdout.

Dataset split example, Fig. 4-2 (1 mark): A grey test set is held out. The rest is split three ways; each hyperparameter combination is trained on blue parts and validated on green parts. The grey set is used once, with the best combination.

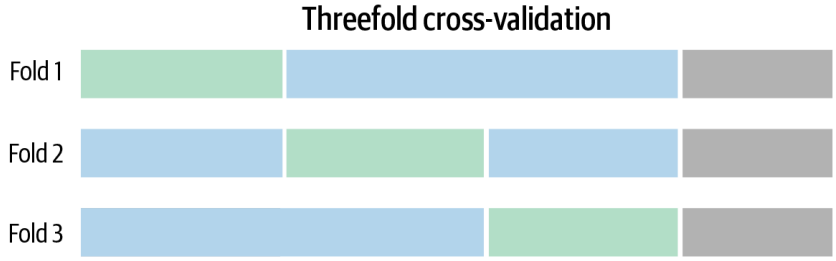


Figure 4-2. An example of dataset split for model evaluation (Source: Treveil et al., Introducing MLOps, 2020)

Q23. Explain the impact of Responsible AI on modelling, with the hiring example on fairness. (8 marks)

Topic: Evaluating and Comparing Models (Impact of Responsible AI on Modeling) · Ch. 4, pp. 53–56

Explainability (2 marks): Some contexts need individual predictions explained, showing which features pushed a prediction. Methods: **Shapley values** (average marginal contribution of a feature value across all coalitions) and **individual conditional expectation (ICE)** (dependence of the target on features of interest). Example: a hormone level usually signals a health issue, but not for a pregnant woman. Some laws require explanations for decisions affecting people, like a loan denial.

Hiring example (4 marks): A US organization predicts worker performance to hire 10 people. The population has two hypothetical groups, Weequay and Togruta; more Weequay attend university, so Weequay applicants are more numerous and more qualified.

- **Equal opportunity:** for each applicant, all else equal, hiring must not depend on group. This yields 60% or more Weequay hires.
- **Disparate impact:** practices must not adversely affect one protected group more than another, judged on subpopulations by hiring proportions.
- The two goals are mutually exclusive: equal opportunity causes disparate impact, and forcing 40% Togruta means rejecting some Weequay predicted more qualified.
- **80% rule:** the Togruta hiring rate must be at least 80% of the Weequay rate, allowing up to 65% Weequay hires.

Implementation problems (2 marks):

- Two separate models (one per group) address over-representation but amount to disparate treatment, possibly discriminatory.
- Post-processing with a higher threshold for Weequay adjusts the trade-off but may still be disparate treatment.
- Objectives cannot be set by data scientists alone; bias is as much a business question as a technical one, and depends on context (granting privileges, negative impact like fraud rejection, or neutral like disease prediction).

Q24. Explain version management and reproducibility of ML models, listing the facets that must be documented. (8 marks)

Topic: Version Management and Reproducibility · Ch. 4, pp. 56–58

Two needs (2 marks):

1. During experimentation, go back to different branches of experiments, e.g., restore a previous project state after a dead end.
2. Let data scientists, auditors or managers replay the computations behind a deployment, possibly years later.

Versioning (1 mark): Largely solved for code via source control; platforms offer similar features for pipelines and model configuration. Merging is harder than for code, but the basic need is returning to an experiment to copy its settings.

Facets to document for reproducibility (5 marks):

1. **Assumptions:** decisions about the problem, scope and data must be explicit and logged.
2. **Randomness:** sampling and many algorithms use pseudo-random numbers; control the generator's seed.
3. **Data:** the same data must be available; versioning data can be storage-heavy and branching tools are immature.
4. **Settings:** all processing must be reproducible with the same settings.
5. **Results:** compare in-depth analyses (confusion matrices, partial dependence plots), not just text diffs.
6. **Implementation:** slightly different implementations can change close-call predictions; bulk and live API scoring may justify different implementations.
7. **Environment:** a different Python package version can change results; freeze computation environments.

Documentation can be partly automated, and an integrated platform lowers reproducibility costs. This is critical where governance and audits matter.

Module 2 – 6 Mark Questions

Q25. Explain the data exploration process carried out before training a model. (6 marks)

Topic: Data Exploration · Ch. 4, p. 46

Purpose (1 mark): Even the best algorithm is only as good as its training data. Exploration reveals issues such as incompleteness, inaccuracy and inconsistency that can make some or all data unusable.

Typical activities (4 marks):

1. Document how the data was collected and the assumptions already made
2. Examine summary statistics: each column's domain, missing values, obvious mistakes, strange outliers (or a suspicious absence of outliers)
3. Look closely at the data's distribution
4. Clean, fill, reshape, filter, clip and sample
5. Check correlations between columns, run statistical tests on subpopulations, fit distribution curves
6. Compare with other data or models in the literature: is usual information missing, is the distribution comparable?

Domain knowledge (1 mark): Some oddities need specialist insight. For industrial sensor data, a data scientist who is not a mechanical engineer may not know which outliers are normal for a machine.

Q26. How does feature selection impact MLOps strategy? Explain automated feature selection and feature stores. (6 marks)

Topic: Feature Engineering and Selection · Ch. 4, pp. 48–49

Trade-off (1 mark): More features may improve accuracy, give fairer finer-grained groups or compensate for missing information.

Downsides for MLOps (2 marks):

- The model becomes more expensive to compute
- More features need more inputs and more maintenance
- More features mean some loss of stability
- A large number of features can raise privacy concerns

Automated feature selection (1 mark): Heuristics estimate feature importance, e.g., correlation with the target or training a quick simple model on a subset to find the strongest predictors. Tools like Auto-sklearn or AutoML cross-reference features against the target and drop those unlikely to help.

Human choices (1 mark): Deciding whether to collect more data still needs people. Business-friendly features improve performance, ease adoption through simpler explanations and reduce modelling debt.

Feature stores (1 mark): Feature factories or stores keep features tied to business entities in a central place for reuse. They combine a slower, powerful offline part and a fast online part for real-time needs, kept consistent. They free data scientists' time and help reduce ML's technical debt.

Q27. Differentiate between high-bias and high-variance models. Explain stratified modelling with an example. (6 marks)

Topic: Experimentation · Ch. 4, p. 50

Aspect	High bias (underfitting)	High variance (overfitting)
Behaviour	Fails to learn rules that the training data could teach	Sees patterns in noise and tries to predict every variation
Cause	Reductive assumptions; model too simple	Model too complex for the signal
Result	Poor fit even on training data	Fits training data but generalizes poorly

Experimentation tunes this bias/variance trade-off for a given training cost (4 marks for the comparison).

Stratified modelling (2 marks): Some tools automate training one model per segment. Example: a business forecasting product demand to optimize inventory finds behaviour varies greatly between stores. Training one model per store optimizes each store better than a single model for all stores.

Q28. Explain the steps taken to cross-check model behaviour beyond raw metrics. (6 marks)

Topic: Evaluating and Comparing Models (Cross-Checking Model Behavior) · Ch. 4, p. 53

Why (1 mark): Understanding behaviour matters in proportion to the model's impact. A model that sends every patient to a doctor scores high on prevention but fails on realistic resource allocation.

Reasonable steps (3 marks):

1. Cross-check several metrics, not only the one optimized
2. Check reactions to different inputs, e.g., plot the average prediction (or probability) for values of an input and look for oddities or extreme variability
3. Split along a dimension and compare behaviour and metrics across subpopulations, e.g., is the error rate the same for males and females?

Correlation, not causality (1 mark): These analyses show how the model sees relationships, not causes. What-if analysis needs care: predictions are likely wrong for feature values, or combinations, unseen in training.

Comparison environment (1 mark): Data scientists need the full environment (tools, data) for all models to compare from every angle: for drift, same settings with different data; for performance, same data with different settings.

Q29. Explain cross-testing, k-fold cross-validation and the dataset split scheme used for model evaluation. (6 marks)

Topic: Evaluating and Comparing Models (Choosing Evaluation Metrics) · Ch. 4, p. 52 (Fig. 4-2)

Cross-testing (2 marks): To estimate generalization, the metric is computed on a holdout dataset not used for training. Data may be held out at several stages, e.g., metric evaluation and hyperparameter optimization.

k-fold cross-validation (1 mark): The held-out part is rotated so the model is trained and evaluated several times. Training time multiplies, but the spread of results shows metric stability.

Recent-records holdout (1 mark): In a simple split, holding out the most recent records mimics future use and gives more realistic estimates. It also reveals drift between training and holdout data.

Fig. 4-2 scheme (2 marks): A grey test set is set aside. The remaining data is split into three parts; each hyperparameter combination is trained three times on the blue datasets and validated on the matching green datasets. The grey set is used only once, with the best combination; the others are used with every combination.

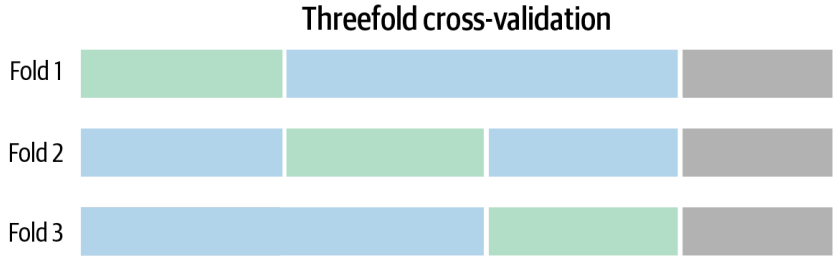


Figure 4-2. An example of dataset split for model evaluation (Source: Treveil et al., Introducing MLOps, 2020)

Q30. Explain explainability of individual predictions and why deep learning models are called black-box models. (6 marks)

Topic: Evaluating and Comparing Models (Impact of Responsible AI on Modeling) · Ch. 4, pp. 53–54

Need (1 mark): Depending on the situation or industry, individual predictions must be explainable, showing which features pushed the prediction one way or the other. A record's prediction can differ greatly from the average.

Methods (2 marks):

- **Shapley value:** the average marginal contribution of a feature value across all possible coalitions
- **Individual conditional expectation (ICE):** shows dependence between the target function and features of interest

Example (1 mark): A high level of a certain hormone usually leads a model to predict a health issue, but for a pregnant woman the model infers no such risk. Some laws require explanations for decisions affecting people, such as recommending a loan denial.

Black-box models (2 marks): Explainability has several dimensions. Deep learning networks are fully specified by their coefficients and are conceptually simple, but the formula is so large it cannot be grasped intuitively. Global and local tools (partial dependence plots, Shapley values) give insight but do not make the model intuitive. Rigorous, intuitive understanding requires limiting model complexity.

Q31. Explain equal opportunity, disparate impact and the 80% rule with an example. (6 marks)

Topic: Evaluating and Comparing Models (Impact of Responsible AI on Modeling) · Ch. 4, pp. 54–55

Scenario (1 mark): A US employer uses a model to hire 10 workers. Two hypothetical groups exist, Weequay and Togruta; more Weequay attend university, so Weequay applicants are more numerous and more qualified.

Equal opportunity (2 marks): For each applicant, all else equal, hiring must not depend on group. The organization is usually not accountable for processes outside its control. Here it leads to 60% or more Weequay hires.

Disparate impact (2 marks): Employment practices must not adversely affect one protected group more than another. It is measured on subpopulations, comparing hiring proportions, usually against applicant proportions. Equal opportunity therefore creates disparate impact, while forcing 40% Togruta hires means rejecting some Weequay predicted more qualified. The two goals are mutually exclusive.

80% rule (1 mark): A legal compromise: the Togruta hiring rate must be at least 80% of the Weequay rate. Here, up to 65% Weequay hires would be acceptable. Setting these objectives is not a decision for data scientists alone.

Q32. Why must a model be evaluated in context? Why is a perfect evaluation score suspicious? (6 marks)

Topic: Evaluating and Comparing Models · Ch. 4, p. 51

“All models are wrong, but some are useful” (2 marks): This idea from statistician George E. P. Box means a model need not be perfect, only useful. Watch for the uncanny valley: a

model that looks good overall but performs badly or catastrophically on a subset, such as an under-represented population.

Evaluating in context (2 marks): Compare the model with what existed before, a previous model or a rule-based process, to see the outcome of replacing it. A model with disappointing absolute performance can still improve the situation; a slightly better demand forecast may bring large cost savings.

Suspicious perfect score (2 marks): Most problems have noisy data that is hard to predict, so a perfect or near-perfect score is a warning sign of:

- **Data leakage:** the target is present in the input data, or a feature is highly correlated with the target but in practice available only once the target is known
- **Overfitting:** the model fits the training data and will not generalize well

Module 3

Preparing for Production & CI/CD Pipelining

Module 3 – 8 Mark Questions

Q33. Explain the runtime environment considerations when preparing an ML model for production. (8 marks)

Topic: Runtime Environments · Ch. 5, pp. 60–63

Variety of production environments (1 mark): Production environments take many forms: custom-built services, data science platforms, dedicated services such as TensorFlow Serving, low-level infrastructure such as Kubernetes clusters, and JVMs on embedded systems. Several heterogeneous environments may coexist in one organization. Ideally, a model validated in development is sent to production as is, which minimizes adaptation and keeps its behaviour unchanged.

Adaptation from development to production (2 marks):

- At one end, development and production platforms are from the same vendor or interoperable, so the model runs unchanged and pushing it takes a few clicks or commands; all effort goes into validation.
- At the other end, the model must be reimplemented from scratch, possibly by another team in another language. This usually reflects a lack of tooling and processes, and the model may not reach production for months or years, if at all.
- In all cases, validation must be done in an environment that mimics production as closely as possible, not in the development environment.

Format and tooling considerations (2 marks): The production format should be decided early because it affects the model and the work needed. For example, a scikit-learn (Python) model sent to a Java environment that expects PMML or ONNX needs conversion. The conversion tooling should be set up while developing the model; otherwise validation is blocked, and final validation must be done on the converted model, not the scikit-learn one.

Data access (2 marks): A model may need data the requester does not supply; e.g., an apartment-price model uses the average market price of a zip code, but the request gives only the zip code, so a lookup is needed. Data can sometimes be frozen and bundled with the model. Otherwise production needs network connectivity, libraries or drivers for the data store, and stored credentials. Technical connectivity is a common source of production malfunction.

Final thoughts (1 mark): Training is the most impressive computation, but most compute over a model's life is spent on inference, since a model is trained once and used billions of times. A useful practice is to build a complex model only as a baseline and put a simpler model in production if it comes close, lowering risk, latency and power consumption.

Q34. Explain model risk evaluation, covering the purpose of model validation and the origins of ML model risk. (8 marks)

Topic: Model Risk Evaluation · Ch. 5, pp. 63–64

Why evaluate risk (1 mark): Models mimic reality imperfectly; their implementation and runtime environment can have bugs, and a seemingly insignificant fault can have tremendous consequences in a complex system.

Purpose of model validation (3 marks): Risks must be anticipated so that design and validation minimize them. Involuntary malfunctions or malicious attacks threaten most enterprise ML uses, not only financial or safety applications. Before production (and from the start of the project), teams should ask the uncomfortable questions:

- What if the model acts in the worst imaginable way?
- What if a user manages to extract the training data or the internal logic of the model?
- What are the financial, business, legal, safety and reputational risks?

For high-risk applications the whole team, especially validation engineers, must know these risks to set the right strictness. ML risk management extends model risk management from banking and insurance, but brings new risks to teams without that experience.

Origins of ML model risk (3 marks): Expected cost can be estimated with ML metrics such as the cost matrix, but much can go wrong beyond it. Risk arises from:

1. Bugs or errors in designing, training or evaluating the model (including data prep)
2. Bugs in the runtime framework, post-processing or conversion, or hidden incompatibilities with the runtime
3. Low-quality training data
4. Large difference between production and training data
5. Expected error rates, but with failures of higher consequence than expected
6. Misuse of the model or misinterpretation of its outputs
7. Adversarial attacks
8. Legal risk (copyright infringement, liability for outputs) and reputational risk (bias, unethical use)

Amplifying factors (1 mark): Broad use of the model, a rapidly changing environment, and complex interactions between models.

Q35. Explain quality assurance for machine learning and the key testing considerations before deployment. (8 marks)

Topic: Quality Assurance for ML; Key Testing Considerations · Ch. 5, pp. 64–66

Quality assurance for ML (3 marks):

- Software has mature QA tools, but QA for data and models is still in its infancy; statistical methods and documentation practices are known but rarely implemented at scale.
- QA does not happen only at final validation; it should accompany every stage of model development, ensuring compliance with processes and with ML and computational performance requirements, in proportion to the risk.

- Validators who did not build the model need ML training to design validation or spot gaps, and the organization must give them authority to report issues or block passage to production.
- QA is also the time to create documentation and validate against organizational guidelines: the origin of all datasets, pretrained models and assets must be known (regulations, copyright). Some organizations allow only whitelisted dependencies for safety.

Key testing considerations (5 marks):

1. **Test data:** apply the model to carefully curated data; some scenarios should be realistic, others deliberately problematic (extreme or missing values).
2. **Metrics:** collect statistical metrics (accuracy, precision, recall) and computational metrics (average latency, 95th-percentile latency).
3. **Assertions:** tests should fail when expectations are not met, e.g., accuracy below 90%, average inference time above 100 ms, or more than 5% of inferences above 200 ms.
4. **Comparison and fairness:** compare with the previous version (champion/challenger, sudden metric drops); split data into subpopulations on a sensitive variable to evaluate fairness, which virtually all models applied to people need.
5. **Stability:** a small change in one feature should give a small change in output; unstable models feel unreliable, and simpler or more regularized models are usually more stable.

Q36. Explain machine learning security, covering adversarial attacks, poisoning and other vulnerabilities. (8 marks)

Topic: Machine Learning Security · Ch. 5, pp. 67–69

ML security (1 mark): A deployed model has the usual software security issues, but ML adds threats where an attacker supplies malicious data to make the model err. Early spam filters scored dictionary words, so spammers used exotic Unicode, typos or images.

Adversarial attacks (3 marks):

- A change to an image that is minor or invisible to humans can drastically change a deep network's prediction, because inference is essentially matrix multiplication and small chosen perturbations cause large output changes.
- Example: small stickers on road signs can make an autonomous car's vision system miss or misclassify them.
- The more the attacker knows, the easier it is; complex models need many queries, brute force or a search model. Simple models like logistic regression are essentially immune, while open-source pretrained deep networks are almost always vulnerable.

Poisoning attacks (2 marks): If attackers access even part of the training data, they gain control of the system. Microsoft's 2016 Twitter chatbot began posting offensive tweets within hours because it adapted to offensive input. In practice, the main concern is data from easily manipulated sources.

Other vulnerabilities (1 mark): Models can be used in undesirable ways, e.g., salespeople advising low-score borrowers to shorten their loans, raising risk (correlation is not causality). Models can also leak training data; a nearest-neighbour salary model can reveal a person's exact income from zip code, age and profession.

Governance (1 mark): Assign responsibilities clearly, provide easy channels to report breaches (such as bug bounty programs) and build safety nets. Security is not an add-on; a system must be designed to be secure from the beginning.

Q37. Explain model risk mitigation with respect to changing environments, interactions between models and model misbehaviour. (8 marks)

Topic: Model Risk Mitigation · Ch. 5, pp. 69–72

Controlled rollout (1 mark): The broader the deployment, the greater the risk. When impact is high, new versions should go out through progressive or canary rollouts: served first to a small share of users, increased slowly while monitoring and gathering human feedback.

Changing environments (3 marks):

- Rapid change multiplies risk and can cause harm before monitoring raises alerts; remediation time may be a critical threat, especially if a new model must be built.
- Monitoring must be reactive enough (weekly distributions may be too slow), and procedures should define thresholds that trigger a degraded mode, from a warning to end users to shutting the system down.
- Frequent small changes can keep a model always slightly off its nominal case; this needs long-term monitoring and reevaluation of operating cost. Until retraining converges, a less complex, more consistent model may be better.

Interactions between models (2 marks): The most challenging source of risk; two models are harder to understand than the sum of each because of their interactions. Complexity stays closer to linear when interactions are absent, and redundant independent processing chains can make decisions robust. Chaining models adds significant complexity and unexpected results; independent parallel chains, each short and explainable, are more sustainable.

Model misbehaviour (2 marks):

- Check inputs in real time against the intervals the model was trained and validated on; reject or warn on out-of-bound values.
- Interval checks may be insufficient (e.g., no recent heavy cars in car-price training data); with many features the curse of dimensionality makes this unavoidable, so anomaly detection is used.
- Check outputs before confirming: thresholds on classifier certainty scores (calibrated with conformal prediction) or predetermined intervals for regression. Safeguards should match the model's risk level.

Q38. Explain CI/CD pipelines for machine learning with an example pipeline. (8 marks)

Topic: Deploying to Production – CI/CD Pipelines · Ch. 6, pp. 73–76

CI/CD (1 mark): Continuous integration and continuous delivery (or deployment) form a philosophy of agile development and a set of practices and tools to release applications more often and faster while controlling quality and risk. They apply equally to ML systems and are a critical part of MLOps strategy.

Example pipeline (3 marks): After developing a model, the data scientist pushes code, metadata and documentation to a central repository, triggering:

1. **Build the model:** build the model artifacts; send them to long-term storage; run basic checks (smoke tests, sanity checks); generate fairness and explainability reports.
2. **Deploy to a test environment:** run tests validating ML and computational performance; validate manually.
3. **Deploy to production:** deploy the model as a canary; then fully deploy it.

Incremental approach (1 mark): There is no one-size-fits-all methodology. Start from a simple but fully functional workflow and add steps as quality or scaling challenges appear, adapting to the organization's specific risks.

Continuous integration and version control (2 marks): CI avoids wasted effort merging contributions and detects bugs and conflicts early. The first step is centralized version control. Git keeps a clear history, allows safe rollback and lets contributors work on branches before merging. Git is not designed for large binaries such as model weights or for data; data versioning uses Git extensions, file formats or databases.

Automation and tooling (1 mark): Automating tests is essential; slow or manual testing discourages frequent deployment and delays the discovery of bugs. Jenkins, the most widespread CI tool, can orchestrate CI/CD pipelines for data science.

Q39. Explain deployment strategies for ML models: categories of deployment and considerations when sending models to production. (8 marks)

Topic: Deployment Strategies · Ch. 6, pp. 77–79

Categories of model deployment (3 marks):

- **Batch scoring:** whole datasets processed by the model, e.g., in daily scheduled jobs. It can be parallelized with a runtime such as Apache Spark or by splitting datasets (partitioning or sharding).
- **Real-time scoring:** one or a few records scored, e.g., scoring a user session to choose which ad to show. Deploying many instances is simpler because records can be dispatched across machines with a load balancer.
- There is a continuum: scoring one record can equal a batch of one, and in both cases multiple model instances raise throughput and may lower latency.

Blue-green deployment (2 marks): The first concern is avoiding downtime, especially for real-time scoring. Instead of shutting down and upgrading, a new system is set up beside the stable one; once functional, the workload is redirected to it and the old one is shut down. Also called red-black deployment; frameworks such as Kubernetes support it natively.

Canary releases (3 marks):

- The stable model stays in production while a percentage of the workload goes to the new model and results are monitored; performance and statistical tests decide whether to switch fully, possibly in increments.
- A malfunction affects only a small portion of the workload ("limit the blast radius"), but requests must be picked carefully: a canary serving one region may miss problems elsewhere. Random selection with user affinity is more robust.
- Canary releases can enable A/B testing (comparison on a business metric), but need advanced tooling for metrics, alerts and results.

Q40. Explain containerization for ML deployment and the role of Docker and Kubernetes. (8 marks)

Topic: Containerization · Ch. 6, pp. 79–81

Need for containerization (2 marks): Managing model versions requires an exact description of the environment (Python libraries and versions, system dependencies). Storing this metadata is not enough; deployment must rebuild the environment automatically on the target machine. Target machines run several models that may have incompatible dependency versions and may compete for resources, so one misbehaving model can hurt co-hosted models.

Containers (2 marks): Containerization bundles an application with its configuration files, libraries and dependencies so it runs across environments. Unlike virtual machines, containers do not duplicate the whole operating system; they share one, so they are far more resource efficient. Docker, open source and released in 2014, is the de facto standard; it packages an application, sends it to a Docker host and runs it in isolation.

Issues a serving framework must solve (2 marks):

- Which Docker host(s) should receive the container?
- How is the workload balanced across several copies of a model?
- How is an unresponsive model (e.g., failed machine) detected and its container reprovisioned?
- How is a model on many machines upgraded so that old and new versions switch correctly and the load balancer is updated in sequence?

Kubernetes (2 marks): The open-source standard for container orchestration, with a declarative API to run applications on a group of Docker hosts (a Kubernetes cluster). Users state the desired state, e.g., "make sure four instances of this container run at all times", and Kubernetes allocates hosts, starts and monitors containers and replaces failed ones. Cloud providers offer managed Kubernetes. Docker with Kubernetes simplifies blue-green and canary deployments and scaling, though it cannot natively analyse ML performance.

Module 3 – 6 Mark Questions

Q41. What is an ML artifact? Explain its contents and the testing pipeline. (6 marks)

Topic: Building ML Artifacts · Ch. 6, pp. 75–76

ML artifact (1 mark): Once code and data are in a central repository, a testable and deployable bundle of the project, called an artifact, is built. It goes through a testing pipeline and is made available for deployment.

Contents (2 marks):

- Code for the model and its preprocessing
- Hyperparameters and configuration
- Training and validation data
- Trained model in its runnable form
- Environment, including libraries with specific versions and environment variables
- Documentation
- Code and data for testing scenarios

Testing pipeline (3 marks): Besides checking requirements, good tests make it easy to diagnose the source of a failure, so naming tests and choosing datasets matters:

1. A **base case** test on a fixed, simple dataset with lenient thresholds runs first; failure suggests a programming error or misuse.
2. Datasets with one specific oddity each (missing values, extreme values, synthetic data) show which kind of data breaks the model and protect against adversarial examples.
3. Tests on **recent production data** from several time windows, analysed automatically once the model is deployed.

Q42. Explain reproducibility and auditability in MLOps. (6 marks)

Topic: Reproducibility and Auditability · Ch. 5, pp. 66–67

Reproducibility (2 marks): In academia it means an experiment is described well enough for another competent person to replicate it from the explanations. In MLOps it also means being able to easily rerun the exact same experiment: the model comes with detailed documentation, the training and test data, and an artifact bundling the implementation with the full environment specification. It is needed to prove findings, debug and build on earlier experiments.

Auditability (3 marks): Adds requirements: the full history of the ML pipeline must be accessible from central, reliable storage, with metadata on all model versions:

- The full documentation
- An artifact that runs the model with its exact initial environment
- Test results, including model explanations and fairness reports
- Detailed model logs and monitoring metadata

Value and cost (1 mark): Auditability can be mandatory in regulated applications but helps all organizations with debugging, continuous improvement and tracking actions and responsibilities. It must allow intuitive human understanding of the system and its versions; full auditability has a cost that should match the model's criticality.

Q43. Explain performance considerations in runtime environments and the model compression techniques. (6 marks)

Topic: Runtime Environments – Performance Considerations · Ch. 5, pp. 61–62

Performance-driven conversion (2 marks): A Python model typically has higher latency for scoring a single record than its C++ equivalent, which may be dozens of times faster (though it can also be slower, depending on many factors). Performance also matters when models must run on low-power devices: deep networks can have billions of parameters, impossible to run on small devices and slow and expensive on standard servers.

Compression techniques (3 marks): An optimized runtime is not enough; the model definition itself must be optimized:

- **Quantization:** train with 32-bit floating-point numbers but run inference at lower precision; less memory, faster, accuracy mostly preserved.
- **Pruning:** remove weights or even entire layers; radical, but some methods preserve accuracy.
- **Distillation:** train a smaller "student" network to mimic a bigger network; done well, it can beat training the small network directly on data.

How they are applied (1 mark): They work best when training is oriented to reduce information loss, so they are not simple post-hoc conversions. They are recent and advanced but common in pretrained NLP models.

Q44. Explain the challenges of scaling ML deployments. Differentiate scalable and elastic systems. (6 marks)

Topic: Scaling Deployments · Ch. 6, pp. 81–83

Two growth challenges (1 mark): Using a model in production on high-scale data, and training ever larger numbers of models.

Scaling data for real-time scoring (1 mark): Trained models are essentially formulas, so Kubernetes can replicate them in as many copies as needed; auto-scaling handles provisioning and load balancing. The main difficulty becomes processing the large volume of monitoring data.

Scalable vs elastic (2 marks):

Scalable (horizontally scalable) system	Elastic system
More computers can be added incrementally to expand processing power (e.g., a Kubernetes cluster grown to hundreds of machines)	Scalable, and resources can also be easily added and removed to match demand

Scalable (horizontally scalable) system	Elastic system
A one-machine system is hard to upgrade; migration may be costly and interrupt service	E.g., cloud auto-scaling adds machines when usage is high and removes them when low, optimizing resource use

Batch scoring at scale (1 mark): Use a distributed framework such as Spark (often run through Kubernetes, which orchestrates containers but knows nothing of data processing), or partition the data so several machines each score a subset of rows.

Scaling the number of models (1 mark): Computationally simpler but hardest for governance: CI/CD must handle many deployments, human verification cannot be systematic, and per-team pipelines are inefficient. Shared practices, standardized pipelines and tools such as Netflix's Spinnaker help.

Q45. Explain the requirements and challenges of deploying models, including the logging requirements. (6 marks)

Topic: Requirements and Challenges · Ch. 6, pp. 83–84

Deployment scenarios (2 marks):

1. One model on one server
2. One model on multiple servers
3. Multiple versions of a model on one server
4. Multiple versions of a model on multiple servers
5. Multiple versions of multiple models on multiple servers

Logging requirements (2 marks): An effective logging system produces centralized datasets for the model designer or ML engineer, usually outside production. It must:

- Access and retrieve scoring logs from multiple servers, for real-time or batch scoring
- Map and aggregate all information per model across servers
- Map and aggregate all information per model version across servers

Challenges (2 marks): In large-scale applications, the number of raw event logs is a problem without preprocessing to filter and aggregate them. Logging streaming data for real-time scoring needs new tooling and significant engineering effort. Because monitoring usually estimates aggregate metrics, saving only a subset of predictions may be acceptable.

Q46. Differentiate integration, delivery, deployment and release in a deployment pipeline. (6 marks)

Topic: CI/CD Pipelines · Ch. 6, p. 77

These terms are often used inconsistently; agreeing on them makes technical and business processes smoother.

Term	Meaning
Integration	Merging a contribution into a central repository (typically a Git feature branch into the main branch) and running more or less complex tests
Delivery	As in continuous delivery (CD): building a fully packaged and validated version of the model, ready to deploy to production
Deployment	Running a new model version on a target infrastructure. Fully automated deployment is a business decision as much as a technical one; continuous delivery is required for continuous deployment but is valuable without it
Release	Directing the production workload to the new version. Deploying a version does not by itself mean it receives traffic; several versions can run at once on production infrastructure

(1½ marks per term.)

Q47. Explain the maintenance measures for a model in production. How are canary releases related to A/B testing? (6 marks)

Topic: Deployment Strategies – Maintenance in Production · Ch. 6, pp. 78–79

Maintenance measures (3 marks):

- **Resource monitoring:** as for any server application, collect IT metrics such as CPU, memory, disk and network usage to detect and troubleshoot issues.
- **Health check:** query the model at a fixed interval (about one minute) and log the results, to check it is online and analyse latency.
- **ML metrics monitoring:** analyse accuracy, compare with another version and detect staleness; heavy to compute, so typically weekly, depending on the application.

Rollback (1 mark): When a malfunction is detected, rolling back to a previous version may be necessary. The rollback procedure must be ready, as automated as possible and tested regularly.

Canary releases and A/B testing (2 marks): A/B testing compares two versions of an application on a business performance metric. A canary release can be used to carry out A/B testing, but the two are not the same and work at different levels of abstraction: A/B testing could also be coded directly into a single version of an application.

Q48. Explain the key ideas for successfully preparing ML models for production, with a neat diagram. (6 marks)

Topic: Preparing for Production · Ch. 5, pp. 59, 72

Context (1 mark): Something that works in the laboratory may not work in the real world; production differs from development and carries greater commercial risk. Preparing for production starts at development: production requirements, security and risk mitigation must be considered while building the model, and MLOps includes a clear validation step before production.

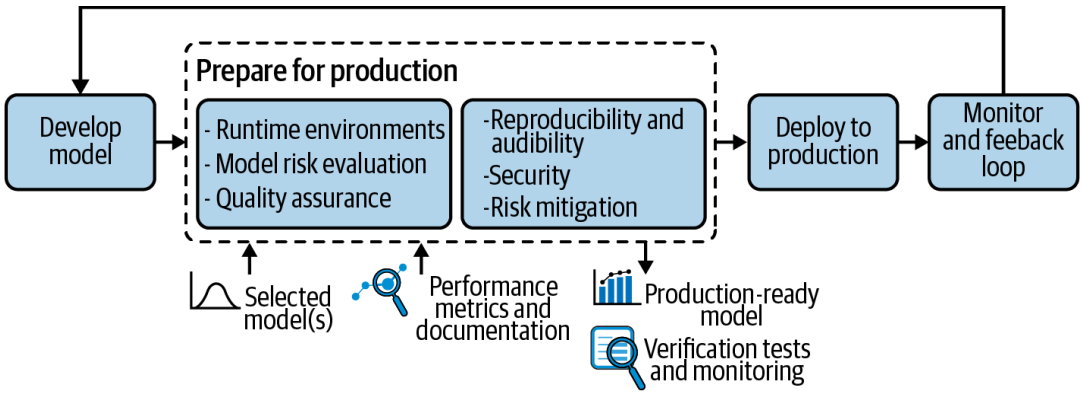


Figure 5-1. Preparing for production highlighted in the larger context of the ML project life cycle (Source: Treveil et al., Introducing MLOps, 2020)

Key ideas (4 marks):

1. Clearly identify the nature of the risks and their magnitudes.
2. Understand model complexity and its impact at multiple levels: increased latency, memory and power consumption, lower ability to interpret inference in production, and harder-to-control risk.
3. Provide a simple but clear standard of quality, ensure the team is appropriately trained, and structure the organization for fast and reliable validation.
4. Automate all validation that can be automated, so it is performed properly and consistently while still deploying quickly.

Diagram (1 mark): Fig. 5-1 — preparing for production (runtime environments, model risk evaluation, quality assurance, reproducibility and auditability, security, risk mitigation) sits between developing the model and deploying it, producing a production-ready model with verification tests and monitoring.

Module 4

Monitoring, Feedback Loop & Governance

Module 4 – 8 Mark Questions

Q49. "Some level of model retraining is necessary — it is not a question of if, but of when." Explain the factors that decide how often models should be retrained. (8 marks)

Topic: Model Retrainment – How Often Should Models Be Retrained? · Ch. 7, pp. 86–89

Factors (3 marks):

- **Domain:** cybersecurity or real-time trading models need frequent updates; physical models such as voice recognition are more stable, but still need to adapt (e.g., a speaker with a cough).
- **Cost:** is retraining worth it? If the pipeline and retraining take one week, is a 1% improvement worth it?
- **Model performance:** if performance is limited by few training examples, retraining depends on collecting enough new data.

Ground truth lag (1 mark): The delay to obtain ground truth sets a lower bound on the retraining period. Using a model that may drift faster than this lag is very risky; a model with a one-year lag is unlikely to be retrained more than a few times a year.

Organizational bounds (2 marks):

- **Upper bound:** retrain at least once a year so the team keeps the skills (despite turnover) and the computing toolchain still works.
- **Lower bound:** even with near-instant feedback (e.g., a recommendation engine), shadow or A/B testing needs time for statistical validation, and human validation or manual rollback makes retraining more than once a day unlikely.

So retraining usually happens between once a day and once a year, typically in the same way and environment as the original training.

Online learning (1 mark): The exception: algorithms that train themselves iteratively. They are costly to set up, must be qualified on changing data, are hard to debug, and are not stateless (running twice on the same data gives different results).

Notifications (1 mark): Organizations need a clear view of drift and accuracy, ideally a pipeline that checks for degradation automatically. Notifications alert the data scientist to diagnose the issue; they need not trigger automatic retraining, since retraining is not always the answer.

Q50. Explain ground truth evaluation for monitoring model degradation and its challenges. (8 marks)

Topic: Understanding Model Degradation – Ground Truth Evaluation · Ch. 7, pp. 89–91

Two approaches (1 mark): Performance degradation is monitored by ground truth evaluation or input drift detection.

Ground truth evaluation (2 marks): It requires waiting for the label event — whether a transaction was actually fraudulent, or whether a customer clicked on or bought a recommended product. The model's performance on the new ground truth is computed and compared with metrics registered at training time; when the difference passes a threshold, the model is outdated and should be retrained. When available, ground truth monitoring is the best solution.

Three main challenges (3 marks):

1. **Not immediately available:** labels may take months or longer, causing economic loss if the model degrades quickly; models with long lags need mitigation measures.
2. **Ground truth and prediction are decoupled:** they are generated and stored in different systems at different times, so matching them is hard. Skipping automated collection for short-lived models is short-sighted, as retraining will be needed.
3. **Only partially available:** labelling every observation can be too expensive, and choosing what to label introduces bias.

Biased labelling and the feedback loop (2 marks): In fraud detection, labelling only cases the model flags creates a feedback loop that amplifies the model's flaws; fraud patterns the model never caught are never used in retraining. Mitigations: randomly label a subsample in addition to flagged cases, or reweight the biased sample to match the population (e.g., low-income applicants). The labelled subset must cover all possible future predictions, even at the cost of some suboptimal decisions.

Q51. Explain input drift with the wine quality example. Describe the common causes of data drift. (8 marks)

Topic: Understanding Model Degradation – Input Drift Detection; Drift Detection in Practice · Ch. 7, pp. 91–93

Wine example (3 marks): The UCI Wine Quality dataset has red and white vinho verde wines with features such as acidity, residual sugar, pH, sulphates and alcohol, and a quality score from 0 to 10; a good wine scores 7 or more. The data is split into wines with alcohol of 11% or above (used to train and score) and below 11% (treated as new incoming data). Quality is unlikely to be independent of alcohol, and alcohol correlates differently with other features in the two sets, so rules learned on one set (e.g., "low residual sugar and high pH means good wine") may be wrong on the other.

Drift (2 marks): The two samples are not identically distributed; ML also needs independence, which breaks with duplicates or when the next sample can be forecast from the previous one. Training on the first set and deploying on the second causes a distribution shift called drift:

- **Feature drift:** alcohol level is a feature of the model, or correlated with its features.
- **Concept drift:** it is not.

Drift detection in practice (1 mark): To react in time, behaviour is monitored on incoming feature values alone, without waiting for ground truth. If distributions (mean, standard

deviation, correlations) diverge from training, performance is likely to change. Retraining on drifted data may not be possible, but mitigations include reverting to a simpler model or reweighting.

Causes of data drift (2 marks):

- **Sample selection bias:** the training sample is not representative, e.g., a discount-effectiveness model where the best discounts go to the best clients. Often caused by the data collection pipeline; the wine set above 11% is a clear case.
- **Non-stationary environment:** source data does not represent the target population, common in time-dependent tasks with seasonality; e.g., wines from one particularly good or bad vintage year.

Q52. Explain the input drift detection techniques and how their results are interpreted. (8 marks)

Topic: Drift Detection in Practice – Input Drift Detection Techniques · Ch. 7, pp. 93–95

Choosing a technique (1 mark): The choice depends on the interpretability needed. Organizations needing proven, explainable methods should prefer univariate statistical tests; if complex multi-feature drift is expected and the black-box effect is acceptable, the domain classifier is a good option.

Univariate statistical tests (3 marks): A test is applied per feature to source and target data, and a warning is raised when results are significant.

- **Kolmogorov–Smirnov test** for continuous features: a nonparametric test of whether two samples come from the same distribution, measuring a distance between empirical distribution functions.
- **Chi-squared test** for categorical features: checks whether observed target frequencies match the expected frequencies from the source data.
- p-values detect drift as quickly as possible, but they show that an effect exists without quantifying it; on large datasets they flag tiny, irrelevant changes, so they should be paired with business-significant metrics (e.g., an average-age drift of a few months).

Domain classifier (2 marks): Stack the original and development datasets and train a classifier to predict each row's origin. Its performance (e.g., accuracy) is the drift score; a high score means new data is distinguishable, so it has drifted. The trained model's feature importance shows which features cause the drift.

Interpretation of results (2 marks): Drift in the target matters because it hits the bottom line directly (e.g., lower credit scores mean fewer loans and lower revenue). Drift in features helps mitigation:

- Reweighting by the feature (e.g., users over 60 rise from 30% to 60%: double their weight and retrain)
- Removing the feature and training a new model without it

Automatic actions on drift are very unlikely; one exception is retraining only when performance drops or significant drift is detected, when deployment is costly.

Q53. Explain the feedback loop in MLOps and the information recorded by the logging system, with a neat diagram. (8 marks)

Topic: Feedback Loop · Ch. 7, pp. 95–97

The feedback loop (2 marks): All effective ML projects have a data feedback loop: information from production flows back to the prototyping environment for improvement. Data from monitoring is sent to model development; if the model works as expected, nothing is done; if performance degrades, an update is triggered automatically or by the data scientist, usually retraining with new labelled data or building a new model with extra features.

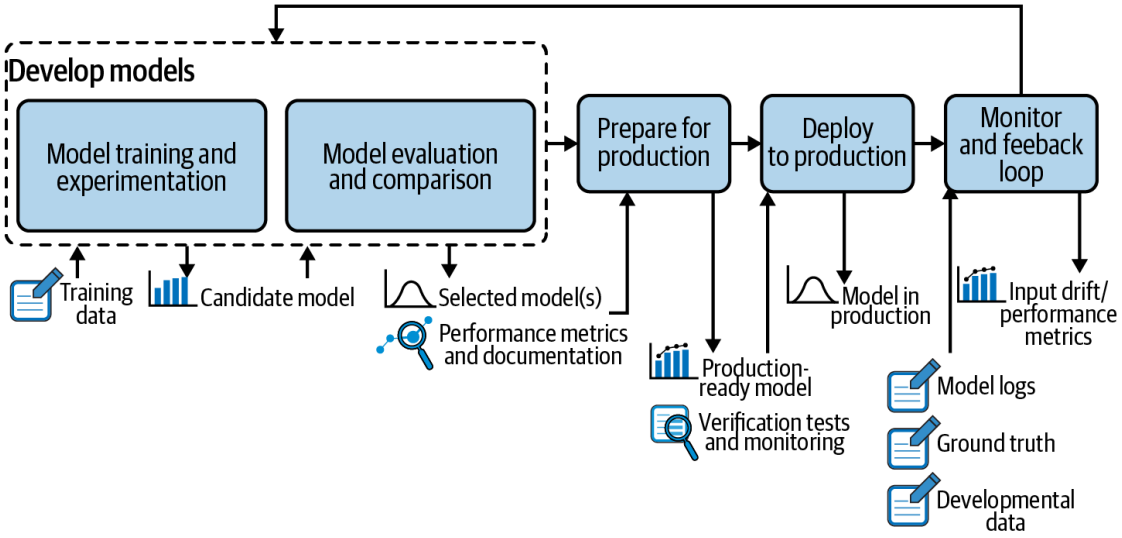


Figure 7-2. Continuous delivery for end-to-end machine learning process (Source: Treveil et al., Introducing MLOps, 2020)

Three components (2 marks):

1. A logging system that collects data from several production servers
2. A model evaluation store that versions models and evaluates different versions
3. An online system that compares models in production, through shadow scoring (champion/challenger) or A/B testing

Logging (3 marks): With several models on several servers, data must be centralized to be analysed. Each event log has a timestamp and:

- **Model metadata:** identification of the model and version
- **Model inputs:** feature values of new observations, to detect data drift
- **Model outputs:** predictions, which with ground truth show production performance
- **System action:** the action taken on the prediction (e.g., blocking a transaction), since it affects user reaction and feedback data
- **Model explanation:** in regulated domains such as finance or healthcare, the features with most influence (e.g., Shapley values), to identify bias or overfitting

Diagram (1 mark): Fig. 7-2 — develop, prepare, deploy and monitor, with model logs, ground truth and input drift/performance metrics feeding back to model development.

Q54. Explain online evaluation of models using the champion/challenger approach. (8 marks)

Topic: Feedback Loop – Online Evaluation · Ch. 7, pp. 99–101

Need (1 mark): Offline and online performance often differ substantially, so testing must move to production for the most truthful feedback. Champion/challenger (shadow testing) and A/B testing both need ground truth; shadow testing should be preferred when possible, as it is simpler to understand and set up and detects differences faster.

Champion/challenger (2 marks): One or more challenger models are deployed to production and score the same requests as the active champion, but return no response; their predictions are only logged. Hence it is also called shadow testing or dark launch.

What it allows (2 marks):

- Verifying that new models perform at least as well as the old one, by direct comparison on the same production data
- Measuring how new models handle realistic load, since new features, preprocessing or algorithms change prediction time
- Giving business leaders visibility of the future champion, reducing the perceived risk of switching

Comparing the models (1 mark): The same information (inputs, outputs, processing time) must be logged for both, so the logging system must tell them apart. They should run until metric fluctuations settle, checked graphically or with a paired-sample t-test; the wider the difference, the fewer predictions are needed.

Concerns (2 marks):

- Two memory-intensive models called synchronously can slow the system and corrupt the collected data.
- Shared external APIs double requests and cost; caching can bias timing comparisons. Scoring only a random subset eases load but takes longer.
- The challenger must not influence the system: its failure must not degrade production, and actions must depend only on the champion and happen once (e.g., never charging a transaction twice).

Q55. Explain how governance should be matched with the risk level of a project, with a neat diagram. (8 marks)

Topic: Model Governance – Matching Governance with Risk Level · Ch. 8, pp. 107–108

Tension between profiles (2 marks): Governance is not a free lunch; it takes effort, discipline and time. Business stakeholders see it slowing delivery and costing money; data

scientists see bureaucracy; risk managers and the DevOps team want strict governance everywhere. MLOps must balance efficiency and protection against threats.

Risk dimensions (2 marks): The balance comes from assessing each project's risk and matching governance to it. Dimensions to consider:

- The audience for the model
- The lifetime of the model and its outcomes
- The impact of the outcomes

The assessment should determine governance measures and drive the whole MLOps development and deployment toolchain.

Examples (2 marks): A self-service analytics (SSA) project for a small internal audience, often built by business analysts, needs lightweight governance. A model on a public website making decisions that affect people's lives or company finances needs a thorough process covering KPIs, algorithm explainability, coding tools, documentation and reproducibility, automated testing, hardware resilience and monitoring. An SSA project with long-term impact can also be high risk, so strategies must be well thought out and reviewed regularly.

Project criticality	Operationalization	Builder autonomy	Versioning	Resources separation	SLA and support by IT	Integration to ext. systems
Irregular Ad-hoc usage	SSA with run on design node	★★★	—	—	—	—
Scheduled but can be inoperative for a small amount of time	Self-service development and scheduling	★★★	★★★	★★	—	—
Scheduled and requires specific monitoring	Light deployment process with rough QA and scheduling	★	★★★	★★★	★	—
Operational projects that cannot suffer outages	Fully controlled deployment CI/CD	—	★★★	★★★	★★★	★★★

Figure 8-1. Choosing the right kind of operationalization model and MLOps features depending on the project's criticality (Source: Treveil et al., Introducing MLOps, 2020)

Diagram (2 marks): Fig. 8-1 — four criticality levels from irregular ad hoc use (SSA on the design node, full builder autonomy) to operational projects that cannot suffer outages (fully controlled CI/CD deployment with full versioning, resource separation, IT SLA and integration with external systems).

Q56. Explain the current regulations driving MLOps governance, with reference to GxP and financial model risk management. (8 marks)

Topic: Current Regulations Driving MLOps Governance · Ch. 8, pp. 108–110

Two forms of regulation (2 marks): Little regulation targets ML and AI specifically, but existing regulations strongly affect ML governance:

- **Industry-specific regulation**, particularly in finance and pharmaceuticals
- **Broad-spectrum regulation**, particularly on data privacy (GDPR, CCPA)

These show what governance will be needed across industry to build trust in ML, even in unregulated sectors.

Pharmaceutical regulation in the US: GxP (3 marks): GxP is a collection of quality guidelines (such as Good Clinical Practice) and regulations from the US FDA to ensure bio and pharmaceutical products are safe. It focuses on:

1. **Traceability:** the ability to re-create the development history of a drug or medical device
2. **Accountability:** who contributed what to development, and when
3. **Data Integrity:** reliability of data used in development and testing, based on the ALCOA principle — attributable, legible, contemporaneous, original and accurate

Financial model risk management (3 marks): Model risk is the risk of losses when models used for decisions on tradable assets (such as Black–Scholes, which predates ML) prove inaccurate. MRM regulation grew from extraordinary events such as crashes, with much added after the 2007–2008 crisis. The UK Prudential Regulation Authority defines four principles:

- **Model definition:** define models and record them in an inventory
- **Risk governance:** a model risk governance framework, policies, procedures and controls
- **Life cycle management:** robust development, implementation and usage processes
- **Effective challenge:** appropriate model validation and independent review

Module 4 – 6 Mark Questions

Q57. Explain the logical model and the model evaluation store with an example. (6 marks)

Topic: Feedback Loop – Model Evaluation · Ch. 7, pp. 97–99

Model evaluation (1 mark): When a drift alert fires and candidates are retrained, all candidates and the deployed model are evaluated on the same dataset. If a candidate is better, the model is updated in production or moved to online evaluation.

Logical model (2 marks): A model changes constantly (retraining on new data, new features, new algorithms), so a higher abstraction is used. A logical model is a collection of model templates and their versions that solves a business problem; a version is obtained by training a template on a dataset. Versions may use different technologies (Python, SQL, Java) but should give the same prediction for the same input.

Wine example (1 mark): Three months after deployment, new data on less alcoholic wine arrives. The result is two templates and three versions: version 1 is live (original template);

version 2 uses the original template on new data; version 3 uses a deep learning template with extra features, trained on the same data as version 2.

Model evaluation store (2 marks): A structure centralizing life cycle data to allow comparisons. Its two tasks:

- **Versioning** the logical model over time, recording features, preprocessing per feature, algorithm and hyperparameters, training dataset, test dataset and evaluation metrics
- **Comparing** versions on the same dataset — preferably new labelled data, otherwise the deployed model's original test set if data has not drifted

Q58. Explain A/B testing for ML models and the MLOps considerations before, during and after the test. (6 marks)

Topic: Feedback Loop – Online Evaluation (A/B Testing) · Ch. 7, pp. 101–103

When to use it (2 marks): A/B testing, a randomized experiment of variants A and B widely used in website optimization, should be used for ML models only when champion/challenger is not possible:

- When ground truth cannot be evaluated for both models (e.g., a recommendation engine cannot know if a customer would have clicked an item not shown; fraud labelling for both models would double the workload)
- When the objective is only indirectly related to prediction (e.g., an ad-click model evaluated on buy rate)

The candidate answers some requests and the original model the rest; statistical tests then compare them.

Considerations (Table 7-1) (4 marks):

Stage	MLOps consideration
Before the test	Define a clear goal (a quantitative business metric, e.g., click-through rate). Define a precise population and an unbiased splitting strategy (e.g., all requests of one customer to one model). Define the statistical protocol: sample size for the minimum effect size and a fixed test duration; power is lower than champion/challenger because unpaired tests are used.
During the test	Do not stop before the planned duration even if results look significant; stopping early (p-hacking) gives unreliable, biased results.
After the test	Check the quality of the collected data, run the statistical tests, and replace the original model if the difference is significant in favour of the candidate.

Q59. Differentiate statistical and business metrics used in ground truth monitoring. Explain the role of p-values. (6 marks)

Topic: Understanding Model Degradation – Ground Truth Evaluation · Ch. 7, pp. 89–90

Statistical metrics (2 marks): Accuracy, ROC AUC, log loss, etc. The metric already used to pick the best model is a first-choice candidate for monitoring; for complex models, metrics by subpopulation may be needed. Their main advantage is that they are domain agnostic, so data scientists are comfortable setting thresholds. Their drawback is that a drop may be statistically significant without noticeable impact, or the cost and risk of retraining and redeploying may exceed the benefit.

Business metrics (2 marks): Cost-benefit assessments; for example, the credit scoring business has developed its own specific metrics. They are far more interesting because they ordinarily have a monetary value, enabling subject matter experts to better handle the cost-benefit trade-off of the retraining decision.

p-values (2 marks): To get the earliest meaningful warning, p-values can assess the probability that an observed drop is not due to random fluctuations. The null hypothesis says there is no relationship between the compared variables (results are due to chance); the alternative says they are related and results are significant. The p-value lies between 0 and 1; the smaller it is, the stronger the evidence to reject the null hypothesis.

Q60. Explain the two levels at which ML models must be monitored. Why can a static model not keep up? (6 marks)

Topic: Monitoring and Feedback Loop · Ch. 7, pp. 85–86

Why monitor (1 mark): A deployed model can degrade fast and without warning until it has harmed the business, so monitoring is a crucial step of the life cycle and a critical piece of MLOps.

Two levels (3 marks):

- **Resource level:** is the model running correctly in production? Is the system alive? Are CPU, RAM, network and disk usage as expected? Are requests processed at the expected rate? This is a traditional DevOps topic.
- **Performance level:** is the model still pertinent over time? Is it still an accurate representation of the pattern of new data? Does it perform as well as during design?

Why static models fall behind (2 marks): Performance reflects how representative the training data is of live request data. The world changes constantly, so a static model cannot catch new patterns without new data. Large deviations on single predictions can be detected directly, but smaller significant deviations must be detected statistically on scored datasets, with or without ground truth. Monitoring tracks this degradation and triggers retraining with more representative data at the right time.

Q61. Who decides what governance an organization needs? Explain the role of regulation, shareholders and public opinion. (6 marks)

Topic: Model Governance · Ch. 8, pp. 105–106

Regulation (1 mark): National regulations safeguard fairness but take time to agree and implement, so they reflect a slightly historical view and cannot anticipate future problems.

Shareholders (1 mark): Businesses want governance to safeguard shareholder investment and a suitable ROI, so the business must perform effectively, profitably and sustainably, with visibility that customers, employees and regulators are satisfied and that future difficulties will be detected and managed.

What is different with ML (1 mark): ML is new, often opaque and risky, yet rapidly embedded in decisions affecting every aspect of life; it invents its own statistically driven, hard-to-understand decision processes based on large volumes of data.

Public opinion (2 marks): It evolves faster than regulation, follows no formal process and need not be based on fact, yet decides what people buy, where they invest and what governments regulate; it decides what is fair. Genetically modified crops were banned in many European countries in the 1990s after public opinion turned. ML trust has been shaken by the Facebook–Cambridge Analytica affair and by unintentionally biased black-box judgments in criminal assessment and recruitment tools.

Implication (1 mark): To reap ML's benefits, businesses must protect public trust through strong MLOps governance: assess risks, set their own fairness values and manage data provenance, transparency, bias, performance and reproducibility.

Q62. Explain the GDPR and CCPA data privacy regulations and the rights they give individuals. (6 marks)

Topic: Current Regulations – GDPR and CCPA · Ch. 8, pp. 110–111

GDPR (2 marks): The EU General Data Protection Regulation, first implemented in 2018, sets guidelines for collecting and processing personal data of people living in the EU. Designed for the internet age, it applies to EU visitors of any website wherever it is based; since few sites exclude EU visitors, it has become a de facto standard.

Rights given to people over their personal data (3 marks):

1. Be informed about data collected or processed
2. Access collected data and understand its processing
3. Correct inaccurate data
4. Be forgotten (have data removed)
5. Restrict processing of personal data
6. Obtain collected data and reuse it elsewhere
7. Object to automated decision-making

CCPA (1 mark): The California Consumer Privacy Act is similar to GDPR in who and what it protects, but its scope, territorial reach and financial penalties are more limited.

Q63. Explain the new wave of AI-specific regulations, with the EU's requirements for trustworthy AI. (6 marks)

Topic: The New Wave of AI-Specific Regulations · Ch. 8, pp. 111–112

The new wave (1 mark): Regulations and guidelines specifically targeting AI (and so all ML) are emerging worldwide, with the EU leading through a framework for trustworthy AI. Its white paper stresses AI's benefits while noting scandals and dangers, and sees a values-based framework as a path to global leadership in the data economy.

Seven key requirements (3 marks):

1. Human agency and oversight
2. Technical robustness and safety
3. Privacy and data governance
4. Transparency
5. Diversity, non-discrimination and fairness
6. Societal and environmental well-being
7. Accountability

Scope and influence (1 mark): The EU approach mainly affects high-risk sectors (healthcare, transportation, energy, parts of the public sector) and is expected to be optional for others. Like GDPR, it is likely to have worldwide influence, and many large organizations may opt in for public trust.

Initiatives worldwide (1 mark): Table 8-1 shows similar routes: the OECD (42 signatories, 5 principles for trustworthy AI), the EU (binding for high-risk activities), Singapore (positive, non-sanction approach), the US (federal guidelines, focus on public trust and fairness), the UK (high-level guidelines) and Australia (detailed guidelines with strong consumer protection).

Q64. Explain the emergence of Responsible AI and its components. (6 marks)

Topic: The Emergence of Responsible AI · Ch. 8, pp. 112–113

Emergence (2 marks): As data science, ML and AI adoption accelerated, a loose consensus emerged under the banner of Responsible AI: developing ML systems that are accountable, sustainable and governable. AI systems should do what they are supposed to, remain reliable over time, and be well controlled and auditable. There is no strict definition or single body driving it, yet it has strongly influenced collective thinking and the EU's trustworthy AI regulators.

Components (Table 8-2) (4 marks):

Intentionality — must have	Accountability — must have
Assurance that models are designed and behave in ways aligned with their purpose	Central control, management and the ability to audit the enterprise AI effort (no shadow IT)
Assurance that data comes from compliant and unbiased sources, with a collaborative approach giving multiple checks on model bias	An overall view of which teams use what data, how and in which models

Intentionality — must have	Accountability — must have
Explainability: results should be explainable by humans, ideally not only those who built the system	Trust that data is reliable and collected per regulation, and a central understanding of which models serve which business process; closely tied to traceability

Both rest on a **human-centered approach**: giving people the tools and training to be aware of and act on both components.

Module 4 – 6 Mark Application Questions (L3)

Q65. An e-commerce company collects personal data from visitors in the European Union to train its recommendation model. Apply the GDPR and CCPA data privacy regulations to this case: identify the rights these visitors hold over their data, and explain what the company's ML pipeline must support when a visitor asks to be forgotten or objects to automated decision-making. (6 marks)

Topic: Current Regulations – GDPR and CCPA · Ch. 8, pp. 110–111 (applies Q62)

Does GDPR apply? (1 mark): Yes. GDPR, implemented in 2018, governs the collection and processing of personal data of people in the EU, and because it was written for the internet age it applies to EU visitors of any website, wherever the website is based. The company must therefore meet its requirements even if it is located outside the EU.

Rights of the visitors (3 marks):

1. Be informed about the data collected or processed
2. Access the collected data and understand its processing
3. Correct inaccurate data
4. Be forgotten (have their data removed)
5. Restrict processing of their personal data
6. Obtain the collected data and reuse it elsewhere
7. Object to automated decision-making

What the pipeline must support (2 marks):

- **Right to be forgotten:** the visitor's data must be removable. This needs data lineage — knowing where each dataset came from and how it is used downstream — so the team can find which training data contains the visitor's records and remove them.
- **Objection to automated decision-making:** the system must be able to stop applying the model's automated decision to that visitor.

If the company also serves California residents, the CCPA protects similar people and data, but its scope, territorial reach and financial penalties are more limited.

Q66. An online retailer has built a new advertisement model that predicts whether a user will click an ad, but the business evaluates it on the buy rate. Justify why A/B testing, and not champion/challenger, must be used to compare it with the current model, and plan the A/B test by stating what must be decided before, during and after the test. (6 marks)

Topic: Feedback Loop – Online Evaluation (A/B Testing) · Ch. 7, pp. 99–103 (applies Q58)

Why A/B testing (2 marks): Champion/challenger (shadow testing) is preferred when possible, but here the objective — the buy rate — is only indirectly related to the prediction of a click. A user can react to only one ad shown by one model, so the reaction to the other model's choice can never be recorded for the same request. Champion/challenger cannot measure this, so A/B testing is the only way: each request is handled by one model, and the two groups are compared.

Before the test (2 marks):

- **Clear goal:** a quantitative business metric to optimize — here the buy rate.
- **Precise population:** choose the segment and an unbiased splitting strategy, e.g., a random split in which all requests of one user go to the same model.
- **Statistical protocol:** fix the sample size for the minimum effect size (the smallest difference worth detecting) and the test duration. Unpaired tests are needed, so power is lower than with champion/challenger.

During the test (1 mark): Do not stop the experiment before the planned duration, even if the test already shows a significant difference; stopping early (p-hacking) gives unreliable, biased results.

After the test (1 mark): Check that the collected data is of good quality, run the statistical tests, and replace the current model only if the difference in buy rate is statistically significant in favour of the new model.

Module 5

MLOps in Practice

Module 5 – 8 Mark Questions

Q67. Explain the business use case and model development for consumer credit risk management. (8 marks)

Topic: Consumer Credit Risk Management – Model Development · Ch. 9, pp. 129–131

Business use case (3 marks): When a consumer asks for a loan, the credit institution decides whether to grant it, informed by scores estimating the probability of repayment. Scores are used at different stages:

- **Prescreen:** a score from a few features quickly discards some applications.
- **Underwriting:** a score using all required information gives a more precise basis for the decision.
- **After underwriting:** scores assess the risk of loans in the portfolio.

Analytics have been used for decades (the FICO score since 1995 in the US). Because they affect revenue and customers' lives, these models are under great scrutiny in a highly regulated environment.

Data and target (2 marks): Historical and current customer information is usually available — how much credit is held, whether the customer has been delinquent — sometimes from credit bureaus directly or as a score. The target is harder: a "bad" outcome is a customer not repaying as expected. Waiting for full repayment or loss charge-off takes too long and reduces reactivity, so trade-offs declare "bad" outcomes from indicators before losses are certain.

Model development (3 marks):

- Historically, credit risk models mix rules ("manual feature engineering") and logistic regression; expert knowledge is vital, and segmentation and studying variable interactions take enormous effort.
- Advanced techniques (two-stage models with offset, Tweedie-based generalized linear models, monotonicity constraints) with financial risk management make the field a playground for actuaries.
- Gradient boosting (e.g., XGBoost) has lowered the cost of good models, but the black-box effect complicates validation. Modelers now use individual explanations such as Shapley values to build trust, a critical component of MLOps. All model types must comply with similar regulations, so credit risk management is a precursor of MLOps.

Q68. Explain the model bias considerations and model validation for consumer credit risk models. (8 marks)

Topic: Consumer Credit Risk Management – Model Bias Considerations · Ch. 9, p. 131

Selection bias (3 marks): The model is used to reject applicants, so the population granted loans is not representative of the applicant population. If a new version is trained without care on the population selected by the previous version, it cannot predict accurately on the rejected

population, which is exactly what is expected of it. This effect is called **cherry-picking**. Special methods such as reweighting based on the applicant population, or calibrating with external data, must be used.

Calibration (2 marks): Models used for risk assessment, not only for granting loans, must produce probabilities, not only yes/no outcomes. Probabilities produced directly by prediction models are usually not accurate. Thresholding for a binary decision is fine, but a monotonous transformation called **calibration** is usually needed to recover "true" probabilities as evaluated on historical data.

Model validation (3 marks):

- Test performance on out-of-sample datasets chosen after (sometimes also before) the training data.
- Investigate performance not only overall but per subpopulation: customer segments based on revenue and, with the rise of Responsible AI, gender or any attribute protected by local regulation.
- Failing to do so can cause serious damage, as Apple learned in 2019 when its credit card was said to be "sexist" against women applying for credit.

Q69. Explain how consumer credit risk models are prepared for and deployed to production. (8 marks)

Topic: Consumer Credit Risk Management – Prepare & Deploy for Production · Ch. 9, pp. 131–133

Documentation (2 marks): Given their impact, validation involves significant work and full documentation of:

- The data used
- The model and the hypotheses made to build it
- The validation methodology and results
- The monitoring methodology

Monitoring methodology (2 marks): It is twofold. Because ground truth arrives late (loan duration plus a few months for late payments), **data drift** must be monitored as well as performance. A recession or a change in commercial policy can shift the applicant population. Data drift is measured per customer segment with distance metrics (Kolmogorov–Smirnov, Wasserstein) and finance-specific ones (population stability index, characteristic stability index). **Performance drift** is assessed on subpopulations with generic (AUC) or specific (Kolmogorov–Smirnov, Gini) metrics.

Independent review (1 mark): Documentation is reviewed by an MRM team in a formal, standalone process. In critical cases the validators rebuild the model from the documentation, sometimes with another technology, to confirm understanding and expose hidden bugs. Such lengthy QA rules out quick fixes and makes the MLOps life cycle slow and deliberate.

Deploy to production (3 marks):

- In large financial organizations, production is separate from design and often uses a different, slowly evolving technical stack.
- Production has mainly supported rules and linear models like logistic regression; some handle PMML or JAR files, and less critical cases may use Docker or integrated data science platforms. Operationalization may range from clicking a button to writing a formula from a Word document.
- Activity logging is essential for monitoring. Automating the feedback loop may be unnecessary if monitoring is once or twice a year, but essential if weekly, as for short-term loans of a few months.

Q70. Explain the rise of recommendation engines and the role of machine learning in them. (8 marks)

Topic: Marketing Recommendation Engines · Ch. 10, pp. 135–136

Introduction (1 mark): Recommendation engines have become very popular over 20 years, from Amazon's first book recommendations to digital shops, advertising, and music and video streaming, while their underlying technologies have evolved.

Human-built recommendations (2 marks): Historically, marketing experts set rules from qualitative and quantitative studies that statically defined the impressions shown to customers with given characteristics (giving rise to the legend that men buying diapers on Thursdays and Saturdays also buy beer).

Bottlenecks of manual engines (2 marks): They wasted significant money because:

1. Rules were hard to build on many customer features, since rule creation was manual.
2. Experiments testing many kinds of impressions were hard to set up.
3. Rules were hard to update when customer behaviour changed.

Role of machine learning (3 marks):

- ML removed rules based on human insight. **Collaborative filtering** dominates: it analyses data on millions of customers and tens of thousands of products to recommend without prior marketing knowledge, by finding what similar customers bought, outperforming traditional strategies in cost and efficiency.
- Because strategies are built automatically, they can be updated regularly and compared with A/B testing or shadow scoring.
- Algorithms may still be combined with business rules, e.g., to avoid the filter bubble, not sell a product in a given area, or avoid unethical associations such as offering alcohol to a recovering alcoholic.

Q71. Explain data preparation and the design and management of experiments for recommendation engines. (8 marks)

Topic: Marketing Recommendation Engines – Data Preparation; Design and Manage Experiments · Ch. 10, pp. 137–138

Customer data (2 marks): Data usually available is:

- Structural information (age, gender, location)
- Historical activities (past views, purchases, searches)
- Current context (current search, viewed product)

All information is aggregated into a fixed-size vector of characteristics, e.g., purchases in the last week or month, views in past periods, increase in spending or views in the last month, and previously seen impressions with the customer's response.

Context and recommendations (2 marks): The recommendation context can be added, e.g., days to summer for seasonal products like above-ground pools, or days to monthly payday. The set of possible recommendations must be defined; the same product may come with different offers communicated differently. The **"do not recommend anything"** option is essential, as not all recommendations help, and some customers may not be entitled to certain recommendations (e.g., by geography).

Choosing the KPI (2 marks): A simple strategy predicts the probability of a click, but this is not adequate for the business: **cannibalization** may occur (showing a low-margin product can prevent purchase of a high-margin one), so sales volume rises while revenue falls. Pushing only the organization's interest can also harm it long term. The KPI and evaluation period must be chosen carefully; revenue over two weeks after the recommendation is common practice.

Experiment design (2 marks): As in A/B testing, control and experimental groups are defined before the experiment by randomly splitting the customer base, ideally with customers not recently in other experiments; in pull settings with many new customers, assignment may be decided on the fly. The larger the expected KPI difference, the smaller the groups and shorter the duration. A two-step approach starts with two equal limited groups, then deploys the new strategy to 10% and raises it progressively.

Q72. Explain the model training and deployment challenges faced by MarketCloud in building recommendation engines for SalesCore. (8 marks)

Topic: Marketing Recommendation Engines – Model Training and Deployment · Ch. 10, pp. 138–141

Scenario (2 marks): MarketCloud is a hypothetical mid-sized global software company in Silicon Valley (about $200 million revenue, a few thousand employees). Its SaaS B2B product SalesCore helps businesses track deals, clear administration and create customized offers. MarketCloud wants SalesCore to suggest products automatically to salespeople during calls, based on each customer's information and past interactions, in a pull (inbound calls) or push (outbound calls) context.

Scalability and customizability (2 marks): Each client company has its own data on its products and customers and does not want to share it. With about four thousand clients, MarketCloud needs four thousand recommendation systems instead of one universal one, and it cannot handcraft them one by one.

Monitoring and retraining strategy (2 marks): Each engine is trained on its client's data, so each is a different model with different performance (e.g., good for a beverage client, poor for a telecom client). Manually watching four thousand models is nearly impossible, so monitoring and retraining on degradation must be automated.

Real-time scoring (1 mark): Clients use SalesCore while on the phone; the negotiation changes every minute, so suggestions (e.g., telecom devices based on the customer's office infrastructure) must come in real time, not after the call.

Turning recommendations on and off (1 mark): Responsible AI says human involvement matters (human-in-command design: it should be possible not to use the AI), and adoption is low if users cannot override AI. Some clients prefer their own intuition, so clients get full control to switch the engine on and off.

Q73. Explain the characteristics of power systems that make consumption forecasting important, with a neat diagram. (8 marks)

Topic: Consumption Forecast – Power Systems · Ch. 11, pp. 147–149

Bulk power systems (2 marks): Predictions at various times and geographical scales are vital to operate a power grid safely. Bulk power systems, also called transmission networks, are the backbone of the grid, mainly made of lines and transformers. Only the largest producers and consumers connect directly; others connect through distribution networks covering the last few kilometres.

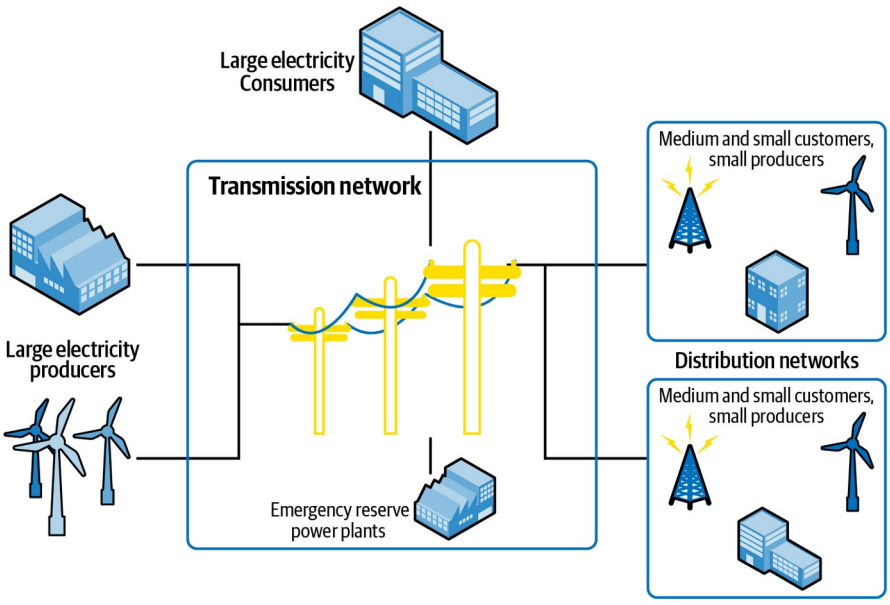


Figure 11-1. A sample bulk power system, to which only the largest producers and consumers are directly connected (Source: Treveil et al., Introducing MLOps, 2020)

Voltage and distance (1 mark): The longer the distance and the larger the energy, the higher the voltage: a few tens of kilovolts for a few tens of megawatts over tens of kilometres, up to one million volts for a few thousand megawatts over thousands of kilometres. One megawatt serves about one thousand inhabitants in Europe.

Two properties needing communication and computation (2 marks):

- **No energy storage:** the network stores less than one second of consumption (up to 30 seconds in alternators and motors), against hours in a gas pipeline, so production and consumption must be balanced very quickly to avoid blackouts.
- **Weak flow control:** unlike telecom networks that drop packets, power grids cannot refuse flow, so a line can exceed its limit; action is needed within seconds to hours of overload.

Anticipation across timescales and geography (2 marks): Operators anticipate contingencies ("if this element fails, is the overload acceptable?") from five minutes to fifty years: below five minutes automatic actions only; five minutes to hours, production schedules and topology; days and seasons, maintenance and contracts; 5–50 years, investment in grid elements lasting decades. Some contingencies have continental impact, so operation requires collecting wide-area data under strong time constraints and processing it to anticipate and act.

Diagram (1 mark): Fig. 11-1 — large producers, large consumers and emergency reserve plants connect to the transmission network, while medium and small customers and producers connect through distribution networks.

Q74. Explain data collection for consumption forecasting using the SCADA and metering systems. (8 marks)

Topic: Consumption Forecast – Data Collection · Ch. 11, pp. 149–151

Two data sources (1 mark): Collecting past data is the first step in forecasting. There are two largely independent sources, the SCADA (supervisory control and data acquisition) system and the metering system; which is used depends on the prediction use case.

SCADA system (3 marks):

- Collects data in real time to give operators an up-to-date view of the system, and sends commands to equipment (e.g., opening and closing a breaker); its synoptic screen is found in most control rooms.
- Refreshes thousands of measurements of flows, consumption and generation every 10 seconds or less.
- Some measurements are intentionally redundant; e.g., power measured at both ends of a line gives the losses, which can be physically estimated to cover missing values, detect anomalies or improve precision.

State estimation and N-1 simulation (2 marks): State estimation uses this redundancy to produce a consistent state of the network, run every few minutes. SCADA raises alarms when an operating limit is not met, but cannot alarm on limits that would be breached if a grid element failed; N-1 simulations of element loss are run regularly on the estimated state. SCADA data loses value quickly, so when stored it is not consolidated (missing values not filled, anomalies not corrected); state estimations are kept for months to a few years.

Metering system (2 marks): Used for invoicing, it need not be as reactive but must be precise. It focuses on generation and consumption, not flow, recording energy withdrawn or injected over periods from a few minutes to one hour. Data used to arrive after a day or more (newer systems in minutes), but consolidation and validation of missing values and anomalies mean final data arrives within a few working days; it is well historized.

Module 5 – 6 Mark Questions

Q75. Differentiate push and pull channels for recommendation engines. (6 marks)

Topic: Marketing Recommendation Engines – Push or Pull? · Ch. 10, pp. 136–137

The structure of a recommendation engine depends on whether recommendations are pushed or pulled.

Aspect	Push channels	Pull channels
Examples	Sending emails, making outbound calls	Online search, calling a customer service line, playlist suggestions on music streaming platforms
Ease	Easiest to handle	Often more effective, as information arrives when customers need it
Execution	Run regularly in batch mode, typically between once a day and once a month	Session information (e.g., what the user searched) tailors the recommendation; prebaked or computed in real time with a special architecture
Experimentation	Easy to split the customer dataset for sound experimental design; regularity allows review and optimization of strategy	Harder: callers differ from the average customer, and strategies may need to stay consistent per customer, giving unbalanced groups and complex statistics
Benefit once set up	Regular review and optimization	Very quick improvement cycle, as many strategies can be tested in real time

(6 marks: 1½ marks per row for any four rows.)

Q76. Why is consumer credit risk management considered a precursor of MLOps? (6 marks)

Topic: Consumer Credit Risk Management · Ch. 9, pp. 129–133

Regulated, scrutinized models (2 marks): Credit scores directly affect institutions' revenues and customers' lives, so these models have always been under great scrutiny; processes, methods and skills have been formalized into a highly regulated environment to ensure sustainable performance.

Same rules for all model types (1 mark): Whether based on expert rules, classical statistics or recent ML algorithms, all models must comply with similar regulations, so parallels and best practices from this use case can be applied elsewhere.

Formal validation (1 mark): Models are fully documented and independently reviewed by an MRM team, sometimes rebuilt from scratch with another technology to confirm understanding and expose hidden bugs.

Adapting to new techniques (1 mark): Financial services have developed validation and monitoring schemes for decades and adapted them to gradient boosting, using explanations such as Shapley values to build trust.

Caution (1 mark): Lengthy QA makes the life cycle slow and deliberate. Other businesses must adapt these practices, as their trade-off between robustness and cost efficiency, time to value and team frustration may differ.

Q77. Explain how the target and scores are defined in consumer credit risk modelling. (6 marks)

Topic: Consumer Credit Risk Management – Background · Ch. 9, pp. 129–130

Scores at each stage (3 marks): The decision to grant a loan is informed by scores estimating the probability of repayment:

1. **Prescreen stage:** a score computed with a small number of features quickly discards some applications.
2. **Underwriting stage:** a score computed with all required information gives a more precise basis for the decision.
3. **After underwriting:** scores assess the risk of loans in the portfolio.

Available information (1 mark): At decision time, information on the customer's history and current situation is available: credit held, and whether the customer has ever been delinquent. In some countries credit bureaus provide this directly or as a score such as FICO (used in the US since 1995).

Defining the target (2 marks): A customer not repaying as expected is a "bad" outcome. In theory, one should wait for complete repayment to confirm a "good" outcome and for the loss charge-off to confirm a "bad" one. This takes too long and reduces reactivity to changing conditions, so trade-offs based on various indicators declare "bad" outcomes before losses are certain.

Q78. Explain the data and performance drift monitoring used for consumer credit risk models. (6 marks)

Topic: Consumer Credit Risk Management – Prepare for Production · Ch. 9, p. 132

Why both are needed (2 marks): The delay between prediction and ground truth is long — typically the loan duration plus a few months for late payments — so monitoring performance alone is not enough; data drift must also be monitored carefully. For example, a

recession or a change of commercial policy could change the applicant population so that performance cannot be guaranteed without further validation.

Data drift (2 marks): Usually measured per customer segment with:

- Generic statistical metrics measuring distances between probability distributions: Kolmogorov–Smirnov or Wasserstein distances
- Metrics specific to financial services: population stability index and characteristic stability index

Performance drift (2 marks): Regularly assessed on subpopulations with generic metrics (AUC) or specific metrics (Kolmogorov–Smirnov, Gini).

Q79. Why is the click probability not an adequate KPI for recommendation engines? How should experiment groups be chosen? (6 marks)

Topic: Marketing Recommendation Engines – Design and Manage Experiments · Ch. 10, p. 138

Need for experiments (1 mark): To use the continuous improvement potential of recommendation engines, strategies must be tested within a sound framework.

Problem with click probability (2 marks): A data scientist may focus on predicting the probability that a customer clicks a recommendation. This seems a reasonable compromise but is not adequate for the business: cannibalization may occur — showing a low-margin product can stop a customer buying a high-margin one — so even good predictions and higher sales volume can reduce overall revenue. Bluntly promoting only the organization's interest can also hurt it long term.

Choosing the KPI (1 mark): The overarching KPI and its evaluation period must be chosen carefully; revenue over a two-week period after the recommendation is common practice.

Choosing groups (2 marks): As in A/B testing, control and experimental groups are defined before the experiment by randomly splitting the customer base, ideally excluding customers in recent experiments. In pull settings with many new customers, assignment can be decided on the fly. Group size and duration depend on the expected KPI difference: the larger the effect, the smaller the groups and the shorter the test. A two-step design starts with two equal limited groups and, if positive, deploys to 10% and raises it progressively.

Q80. Differentiate the SCADA system and the metering system used in power grids. (6 marks)

Topic: Consumption Forecast – Data Collection · Ch. 11, pp. 149–151

Aspect	SCADA system	Metering system
Purpose	Real-time, up-to-date view of the system for the operator; sends commands to equipment (e.g., breakers)	Invoicing

Aspect	SCADA system	Metering system
Requirement	Must be reactive	Need not be as reactive, but must be precise
What it measures	Flows, consumption and generation, thousands of measurements refreshed every 10 seconds or less	Generation and consumption, not flow: energy withdrawn or injected over a few minutes to one hour
Data quality	Some measurements intentionally redundant (e.g., line losses), used by state estimation every few minutes	Missing measurements and anomalies are consolidated and validated
Availability	Immediate, but value fades quickly	Previously after a day or more, now within minutes; final data within a few working days
Storage	Not consolidated when stored; state estimations kept months to a few years	Well historized

(1 mark per row.)

Q81. Explain the two properties of power transmission systems and how operators anticipate contingencies. (6 marks)

Topic: Consumption Forecast – Power Systems · Ch. 11, pp. 148–149

No energy storage (1½ marks): The network stores a meaningless amount of energy: less than one second of consumption in the grid and up to 30 seconds in alternators and motors, compared with several hours in a gas pipeline. Production and consumption must therefore be balanced very quickly to avoid blackouts.

Weak flow control (1½ marks): Telecom networks handle congestion by dropping packets or refusing calls; power grids have no equivalent, so flow on an element can exceed its operating limit, and action is needed within a few seconds to a few hours. Flow control technologies exist, but trade off against instantaneous balance, since power must find a path from generation to consumption.

Anticipating contingencies (3 marks): The operator always asks: if this grid element fails, will the overload on the rest stay acceptable? Actions depend on the horizon:

- Below five minutes: no human action possible; automatic actions must be predefined.
- Five minutes to a few hours: production schedules and grid topology (breakers, flow control).
- A few days ahead: maintenance schedule adaptations.
- A few seasons ahead: maintenance schedules and contracts guaranteeing or limiting capacity.
- 5 to 50 years: investment in grid elements with life expectancies of several decades.

Q82. Explain the real-time scoring requirement and the ability to turn recommendations on and off in MarketCloud's SalesCore. (6 marks)

Topic: Marketing Recommendation Engines – Model Training and Deployment · Ch. 10, pp. 140–141

Real-time scoring (3 marks): MarketCloud's clients often use SalesCore while talking to customers on the phone. The negotiation evolves every minute and the salesperson must adjust the strategy during the call, so the engine must respond to real-time requests. For example, a salesperson selling telecom devices enters the customer's office details (optic fibre, type of WiFi network) and needs suggestions of products the customer could buy immediately — not ten minutes later, after the call, or the next day.

Turning recommendations on and off (3 marks):

- Responsible AI principles recognize that retaining human involvement is important, through a human-in-command design in which it should be possible not to use the AI.
- Adoption is likely to be low if users cannot override AI recommendations.
- Some clients value their own intuition about which products to recommend, so MarketCloud gives clients full control to turn the engine on and off and use recommendations when they want.

Module 5 – 6 Mark Application Questions (L3)

Q83. A bank's personal-loan scoring model, used for loans of three years, was trained before an economic recession. Apply the monitoring methodology for consumer credit risk models to plan how the bank should check whether the model is still reliable, naming the data drift and performance drift metrics it should use. (6 marks)

Topic: Consumer Credit Risk Management – Prepare & Deploy for Production · Ch. 9, pp. 131–133 (applies Q78)

Why performance monitoring alone is not enough (2 marks): Ground truth for a three-year loan arrives only after the loan duration plus a few months for late payments, so a drop in performance would be seen years too late. A recession is exactly the kind of event that changes the applicant population, so the model's performance cannot be guaranteed without further validation. The bank must therefore monitor data drift as well as performance drift.

Data drift (2 marks): Measure drift per customer segment, comparing current applicants with the training population, using:

- Generic distances between probability distributions: Kolmogorov–Smirnov or Wasserstein distance
- Finance-specific metrics: population stability index and characteristic stability index

Performance drift (1 mark): As ground truth arrives, assess performance on subpopulations with generic metrics (AUC) or specific metrics (Kolmogorov–Smirnov, Gini).

Logging and frequency (1 mark): Activity logging of the deployed model is essential. If the bank reviews the model only once or twice a year, the feedback loop need not be automated; if it reviews weekly, automation becomes essential.

Q84. An online electronics store's recommendation engine predicts the probability that a customer clicks a recommendation. After launch, sales volume rose but overall revenue fell. Apply the textbook's guidance to explain the likely cause, choose a suitable KPI with its evaluation period, and design the control and experimental groups for testing a new strategy. (6 marks)

Topic: Marketing Recommendation Engines – Design and Manage Experiments · Ch. 10, p. 138 (applies Q79)

Likely cause (2 marks): Predicting the click probability is not an adequate objective from a business perspective. The likely cause is cannibalization: by showing a low-margin product (e.g., an accessory), the engine can stop the customer buying a high-margin product (e.g., the laptop they came for). Even with good predictions and higher sales volume, overall revenue falls. At the same time, bluntly promoting only the store's interest rather than the customer's can also harm it in the long term.

KPI (1 mark): Choose an overarching KPI and its evaluation period carefully; common practice is the revenue over a two-week period after the recommendation.

Experiment groups (3 marks):

- Define the control and experimental groups before the experiment by randomly splitting the customer base, as in A/B testing.
- If possible, exclude customers who were recently in another experiment, so their history is not polluted; for new customers arriving online, assign them on the fly.
- Size the groups and set the duration from the expected KPI difference: the larger the expected effect, the smaller the groups and the shorter the test.
- Run it in two steps: first two groups of equal but limited size; if positive, deploy the new strategy to 10% of customers and raise the share progressively if results meet expectations.